

5 FUNCTIONS AND OPERATIONS OF RCU

5.1 Summary

This chapter describes procedures for setting parameters and functions for daily operation.



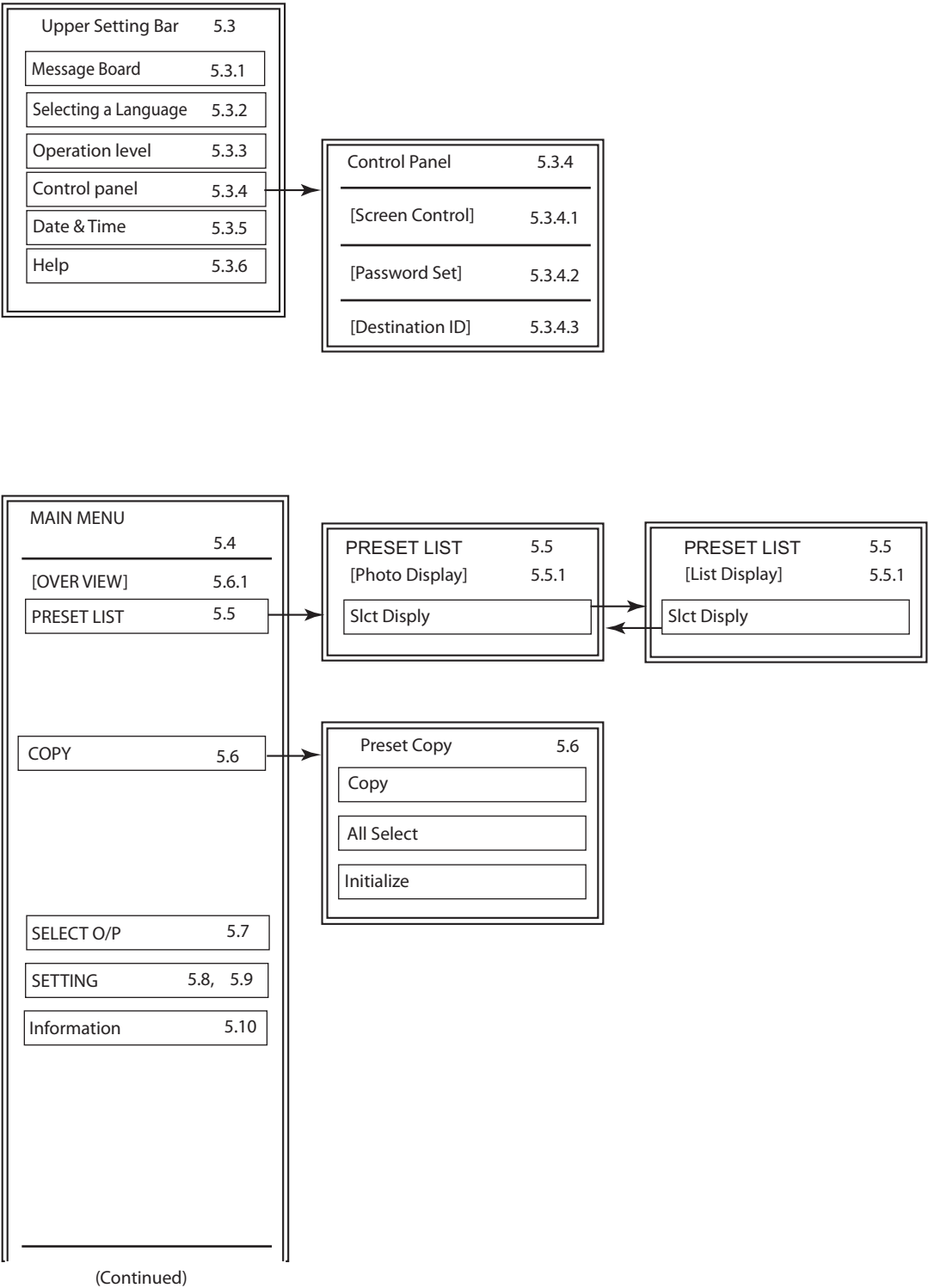
- Be sure to read this chapter carefully before attempting operation.



- Please refer to Chapter 6 for more information on daily production procedures.
- Before operating the RCU unit, please read Section 4.2.

5.2 Screen Configuration Chart

The chart on the following page shows the main TSC-R screens and their access routes. The method of access to each screen is indicated by an arrow from the key to the screen. Please note that some of these screens vary slightly from the actual screens shown on your RCU.



 shows a key.
 shows a screen.

Fig.5-1 Screen Configuration Chart (1)

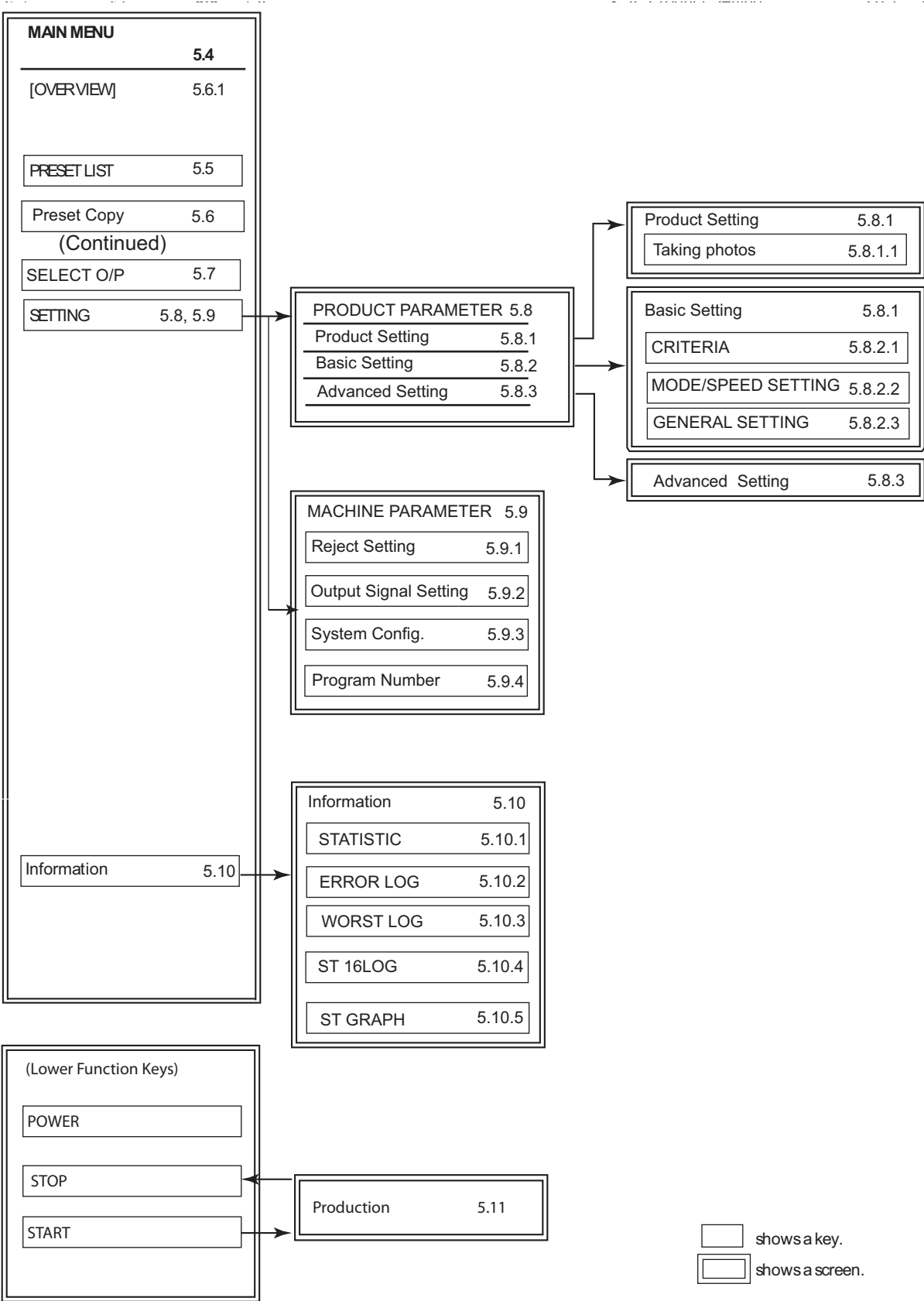


Fig.5-2 Screen Configuration Chart (2)

5.3 Upper Setting Bar

The upper setting bar is displayed at the upper part of the screen, and it is able to set the basic operating environment.



Fig.5-3 Upper Setting Bar (EX: Main Menu Screen)

Table 5-1 Operating Keys at Upper Setting Bar

No.	Name	Display	Function	Reference
1	Message Board		Displays the Message Board.	5.3.1
2	Information display key		Displays Instruction Manual and other information of this thickness and checker.	—
3	Startup assistant key		Not used.	—
4	Language select key		Selects a language.	5.3.2
5	Operation level select key		Switches the operation level.	5.3.3
6	Control panel key		Sets the screen control, password, and data output location.	5.3.4
7	DATE & TIME key		Sets the date and the time.	5.3.5
8	Help key		Displays descriptions of each key. The key blinks by pressing the HELP key. Then, when pressing a key, the description of the key function is displayed. The HELP is closed when touching the screen while HELP is displayed.	5.3.6

5.3.1 Message Board Screen

The message board is the function to handwrite free notes on the operation panel screen.

Any memos and messages, such as a memo about operation or a message to the operator who is taking the next turn, can be freely written down.

Once memo is entered, the message board key blinks. The key blinks even while displaying the other screens, and the presence or absence of the memo data can be recognized at a glance.



CAUTION

- **Be sure to use your fingertip to enter data. The operation panel will be damaged when using a pen or a pointed tool.**

NOTE

- The display may vary depending on the operation speed or writing pressure. Small letters may not be displayed clearly.

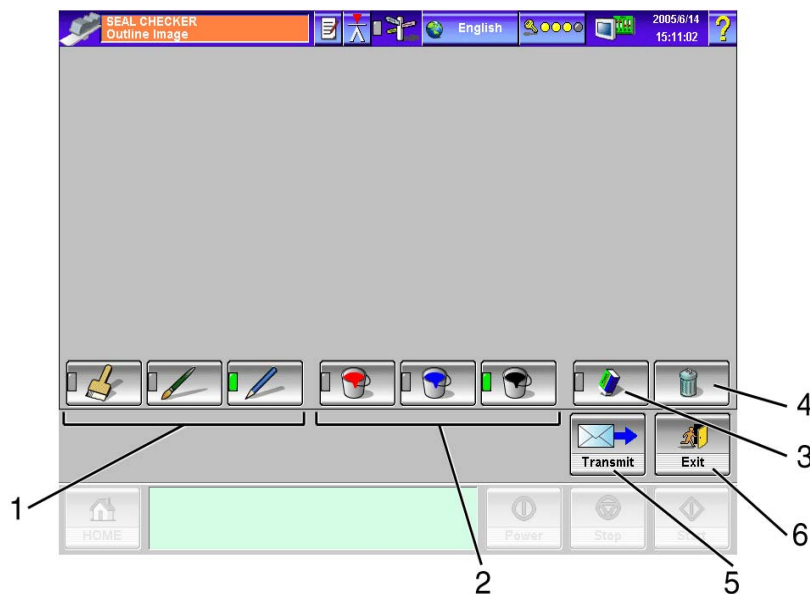


Fig.5-4 Message Board Screen

Table 5-2 Names and Functions of the Keys on the Message Board Screen

No.	Name	Display	Function
1	Line thickness key	  	Selects the line thickness.  : Thick  : Medium  : Thin
2	Line color key		Selects the line color (red, blue, black).

Table 5-2 Names and Functions of the Keys on the Message Board Screen (Continued)

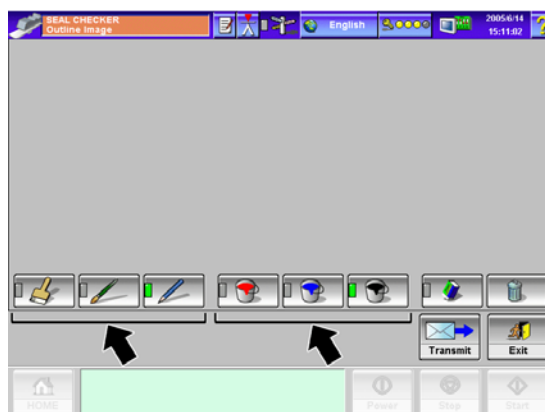
No.	Name	Display	Function
3	Eraser key		Erase the parts followed with this key.
4	Erase All key		Erase the entire displayed memo data.
5	Transmit key		Transmits the message board by e-mail.
6	Exit key		Returns to the previous screen.

< Usage >

1. Press the message board key  on a screen.
2. The Message Board screen will appear.

**Fig.5-5 Main Menu Screen**

3. Selects the line type and line color by each select key.
4. Enter letters with your fingertip.
 - ▶ The entered letters are displayed.
 - ▶ The Message memo key  blinks.

**Fig.5-6 Message Memo Screen**

< When erasing a part of a message >

5. Press the Eraser key .

6. Trace the part to be erased.

▶ The traced part is erased.

NOTE

- Even if the entire message is erased by the eraser key, the memo data cannot be erased entirely and the message board key cannot be released from blinking. The memo data can be erased entirely by using the erase all key, and the message board key can be released from blinking. (See the next page)



Fig.5-7 Message Memo Screen

< When erasing an entire message >

7. Press the Erase All key .

▶ The Confirmation Message will appear.

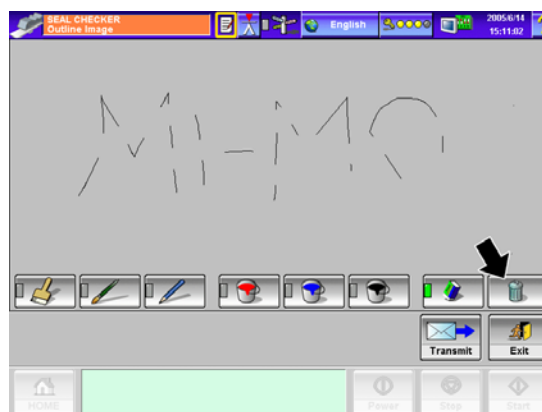


Fig.5-8 Message Memo Screen

8. When not erasing the message, press the No key .

▶ The command will be cancelled.

9. When erasing the message, press the Yes key .

▶ The message board will be entirely erased.

▶ The Message board key will be released from blinking.



Fig.5-9 Confirmation Message to delete

10. Press the Exit key  to end the Message Memo screen.

►The previous screen will appear.

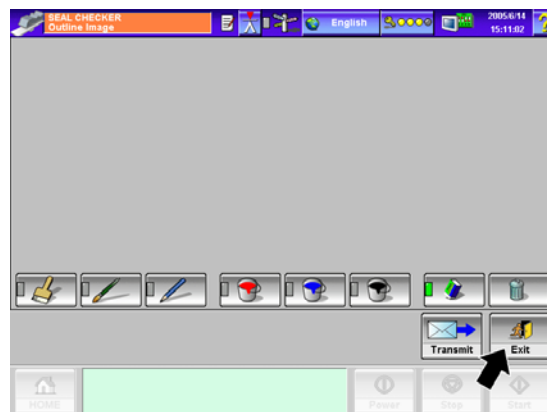


Fig.5-10 Message Memo Screen

< Transmitting the Message Board by e-mail. >

The Message Board can be sent by attaching as an image data in e-mail.

The Mail destination can be set at "E-mail Setting" under "Destination ID" at control panel. (The setting is made under the Maintenance level.)

1. Press Transmit key  on the Message Board screen.

►The Confirmation Message will appear.



Fig.5-11 Message Board Screen

2. When not sending the message, press the No

key .

When sending the message, press the Yes key



►The message board will be transmitted.



Fig.5-12 Confirmation Message (to transmit mail)

5.3.2 Selecting a Language

Select a language on the operation panel.

1. Press the language select key  on a screen.
 - ▶ The language select dropdown list will appear.



Fig.5-13 Main Menu Screen

2. Press the language to be selected.
 - ▶ The selected language key will light up.
 - ▶ The operation language will be switched.



Fig.5-14 Language Select Menu

5.3.3 Selecting an Operation Level

Press the operation level key  on a screen to display the menu to select an operation level.

The operation levels are **Operator**, **Site Engineer**, **Installation**, and **Maintenance**. The operable items differ depending on the operation level.

This section explains operation levels except "Maintenance Level" in the "Table 5-3 Operation Level List."

Table 5-3 Operation Level List

Name	Operation Objected Person	Operation details	Password	
Operation Level	Operator	Basic operation for daily product line work	Unnecessary	Lower ↑
Site Engineer Level	System managers	In addition to the operator level operation, registration for weighing and adjustment operation	Necessary	
Installation Level	Mainly our engineering service person	operation, operations such as adjustment necessary to the equipment installation	Necessary	Upper ↓

The default of operation panel after turning the power on is the operation level.

Changing the operation level to the upper level needs the upper level's passwords.

Passwords have been set as shown in below table at shipment. For changing the password, refer to "Password Setting" (5.3.4.2) in Control Panel.

Entering password is not necessary for changing the upper operation level to the lower operation level.

Table 5-4 Password at shipment

Operation Level	Password
Site Engineer Level	1
Installation Level	2



CAUTION

- Passwords of the higher operation levels than Site Engineer level is for each objected operator for operation.
Keep the passwords secret from non-objected persons.

For changing procedures of operation levels, refer to next pages.

5.3.3.1 Switching to the Operator Level

1. Press the operation level key  on a screen.
 ► The screen to select the operation level will appear.



Fig.5-15 Main Menu Screen

2. Select the Operator level.
 ► The operation level is switched to the the Operator level.



Fig.5-16 Operation Level Selection Menu

NOTE

- Even while displaying the screen for upper levels, the operation level can be switched to the Operator level.
 When switching the operation level to the Operator level while displaying other than the Main Menu screen, the Main Menu screen for the Operator level will be displayed.

5.3.3.2 Switching to the Site Engineer Level

When switching the operation level to the Site Engineer level, the password for the Site Engineer level is required.

- Press the operation level key  on a screen.
 ► The screen to select the operation level will appear.

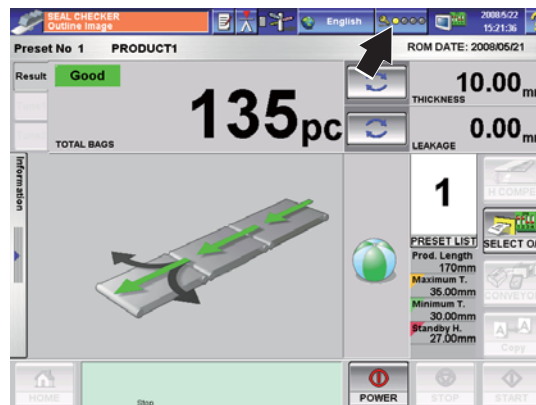


Fig.5-17 Operation Standby Screen

- Select the Site Engineer level.
 ► The keyboard to enter the password will appear.
- Enter the password for the Site Engineer level.
 ► The Main Menu screen for the Site Engineer level will appear.

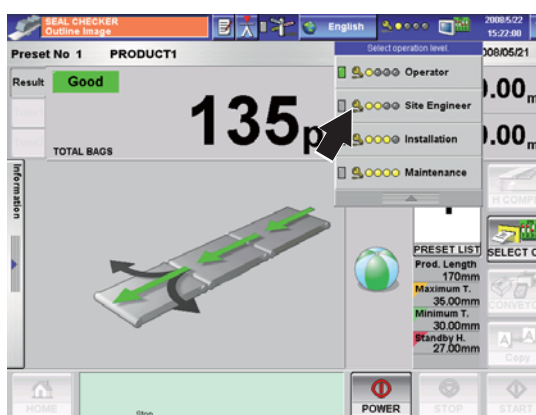


Fig.5-18 Operation Level Selection Menu

5.3.3.3 Switching to the Installation Level

When switching the operation level to the Installation level, the password for the Installation level is required.

Follow the procedure to switch to the Installation level.

- Press the operation level select key  on a screen.
 ► The screen to select the operation level will appear.



Fig.5-19 Main Menu Screen

2. Select the Installation level.
 - ▶ The keyboard to enter the password will appear.
3. Enter the password for the Installation level.
 - ▶ The Main Menu screen for the Installation level will appear.



Fig.5-20 Operation Level Selection Menu

5.3.4 Control Panel

Make settings such as the screen control, password, and data output location on the control panel.

NOTE

- All functions on the control panel can be operated at the Installation level or above.

- Press the control panel key  on a screen.

►The Control Panel menu will appear.



Fig.5-21 Main Menu Screen

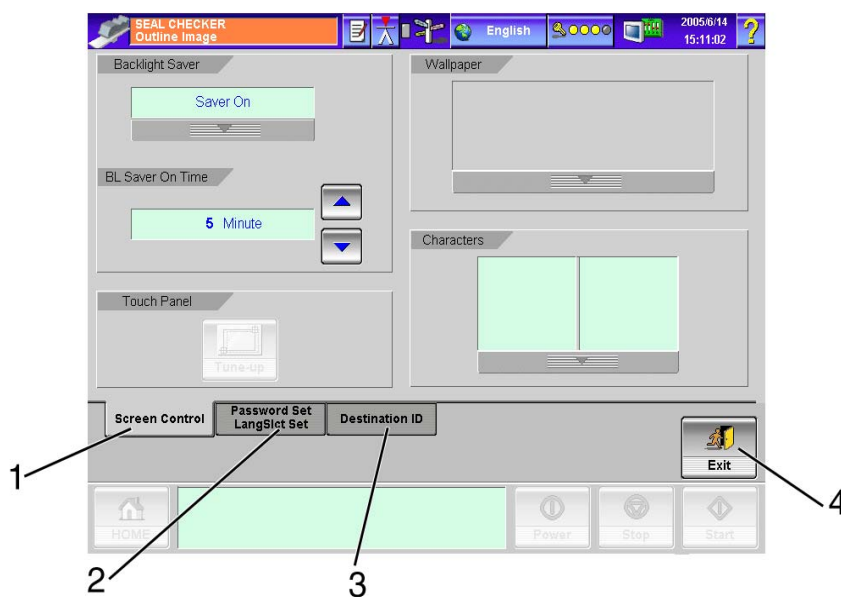


Fig.5-22 Control Panel Menu

Table 5-5 Names and Functions of the Control Panel Screen

No.	Name	Function	Reference
1	Screen Control	Changes the screen control on the operation panel.	5.3.4.1
2	Password Setting, Language Selection Setting	Changes the password of the operation level. Selects an operation language to be displayed on the Language Selection Menu.	5.3.4.2

Table 5-5 Names and Functions of the Control Panel Screen (Continued)

No.	Name	Function	Reference
3	Destination ID	Sets the data destination, access address, and e-mail.	5.3.4.3
4	Exit key 	Terminate the Control Panel screen, and returns to the previous screen.	—

5.3.4.1 Screen Control

The Screen Control screen sets and changes the Screen Control on the Operation Panel.

1. Select the Screen Control tab on the Control Panel screen.
 ► The screen Control screen will appear.
2. Make necessary settings by following the table shown next.

3. Select other Settings Screen or press the Exit

key  to end the setting.

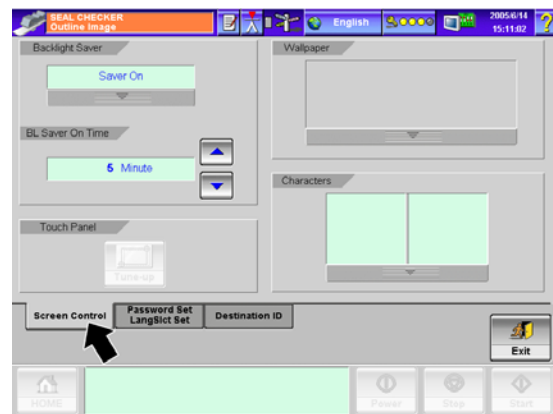
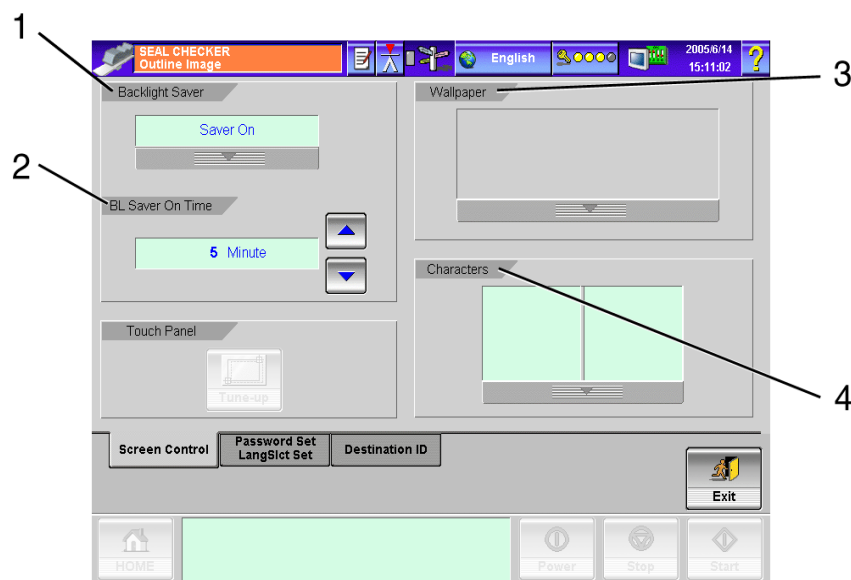
**Fig.5-23 Screen Control Menu****Fig.5-24 Screen Control Menu**

Table 5-6 Setting Item and Functions of Screen Control Menu

No.	Name	Function
1	Backlight Saver	Selects the backlight saver mode (Saver On, Semi brightness, Full brightness). Saver On: After the time set on the Screen Control Menu passes, brightness of the screen changes to semi brightness. Full brightness: Always displayed with full brightness. Semi brightness: Always displayed with semi brightness.
2	BL Saver On Time	When setting at " Save on," sets the time from the last operation of the screen to the semi brightness status of the back light. Setting is available within 0 to 255 minutes by the minute.
3	Wallpaper	Selects the screen wallpaper.
4	Characters	Selects the character for the message window. The expression on the character face tells the unit status.

NOTE

- When the room temperature is 5 °C or lower, the life time of the backlight may be shorten.
It is recommended to use backlight "On" under a cool-temperature environment.

5.3.4.2 Password Set

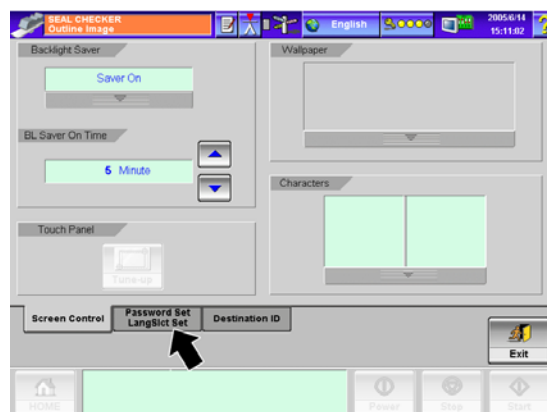
The password for the each operation level can be set and changed on the Password Set screen.

NOTE

- " Language Selection Setting" on the screen cannot be operated under the Installation level.

- Select the Password Set tab on the Control Panel screen.

- " Password Setting" and " Language Selection Setting" screen appears.

**Fig.5-25 Control Panel Menu**

2. Selects the operation level.

►The keyboard will appear.

3. Enter the password and press the return key



4. Select other Settings Screen or press the Exit



key to end the setting.



Fig.5-26 Password Setting, Language Selection Settings Screen

5.3.4.3 Destination ID

The destinations for preset detail and summary detail can be set and changed on the Destination ID screen.

Website address and e-mail address to be displayed on the Information screen can be set.

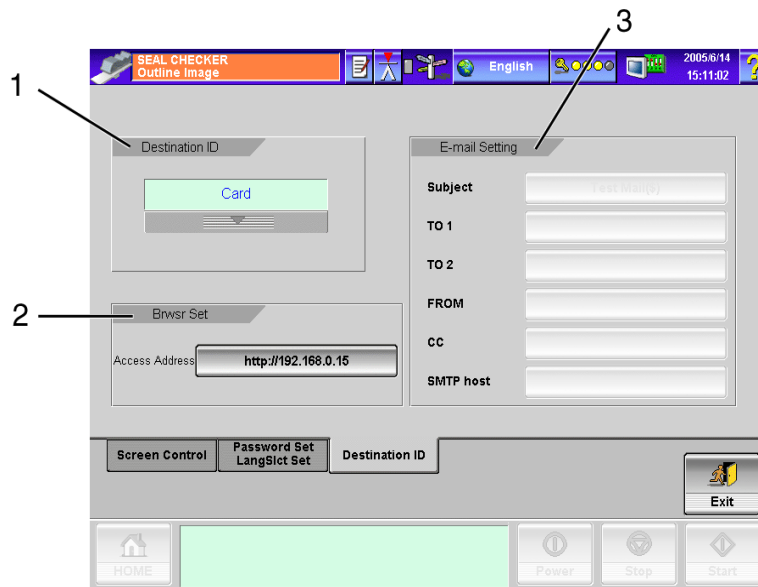


Fig.5-27 Destination ID Menu

Table 5-7 Names and Functions of the Keys on the Destination ID

No.	Name	Function
1	Destination ID	Selects the Destination ID whether or not to output to Printer, File, or E-mail. When the Print key is pressed on the other screen, data is output by the selected output method.
2	Access Address	Enters the website address displayed when pressing the Information display key via the keyboard.

< Setting the Destination ID >

1. Select the Destination ID tab on the Control Panel screen.

► The Destination ID screen will appear.

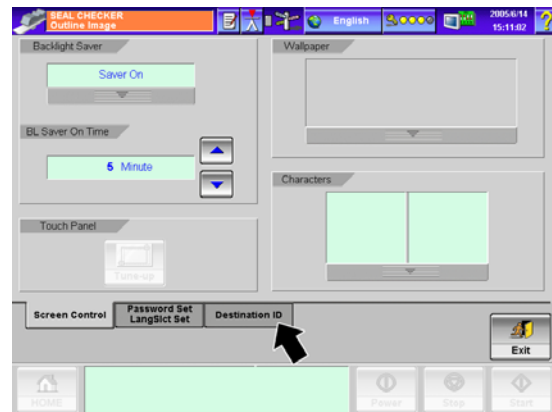


Fig.5-28 CONTROL PANEL Menu

2. Press the drop down key  to select the Destination ID (printer, file, e-mail). Select "NO" when not outputting to any location.

3. Press the Exit key  to end the setting.

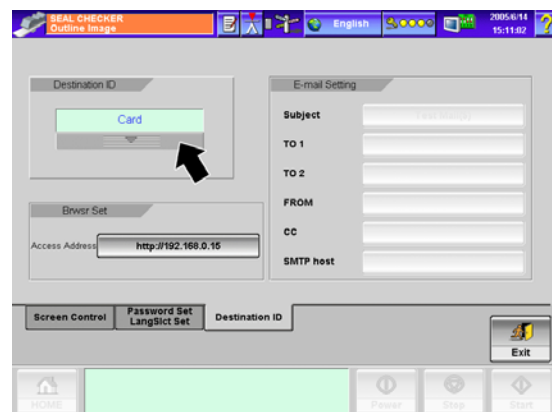


Fig.5-29 Destination ID Menu

< Setting the access address >

1. Press the Access Address key on the Destination ID screen.

► The keyboard will appear.

2. Enter the internet address.

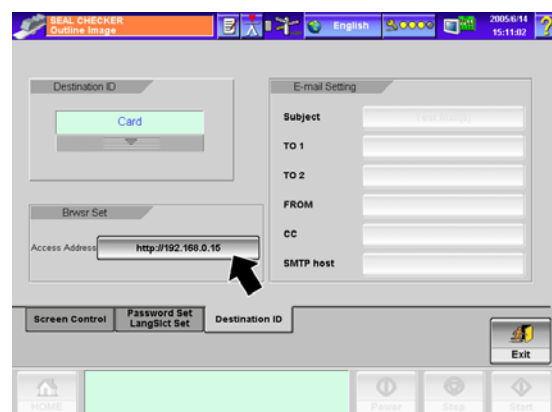


Fig.5-30 Destination ID

Handling the memory card

This weigher is equipped with the memory card slot for saving the preset data and outputting the history.

When saving the output data to the memory card, or reading the reservation data from the memory card, writing to the memory card, and copying between the card and memory, insert the memory card into the card slot.

When setting the data output location as "file", the selected and output data is stored in the memory card.

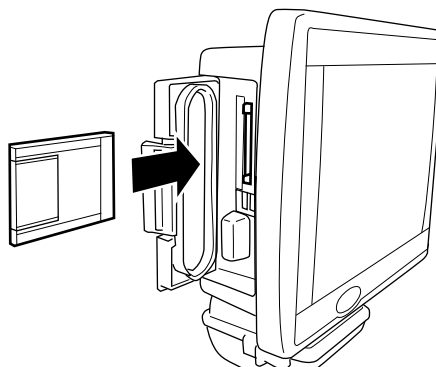


Fig.5-31 Card Slot

CAUTION

- **Read the card with this unit, and do not pull out the card during writing (during accessing). Otherwise, the memory data, or card data may be lost or damaged.**

NOTE

- Make sure to use the memory card that we specified. The other products cannot be used.

5.3.5 Setting the Date & Time

The date and time for the operation panel can be set and changed on the Time Settings screen.

NOTE

- Date & Time can be set at Installation level or over.

1. Press the Time Setting key  on a screen.

► The Time Settings Screen will appear.



Fig.5-32 Main Menu Screen

2. Enter the year and date with Up and Down keys .

TIP

It is possible to select the date on the calendar.

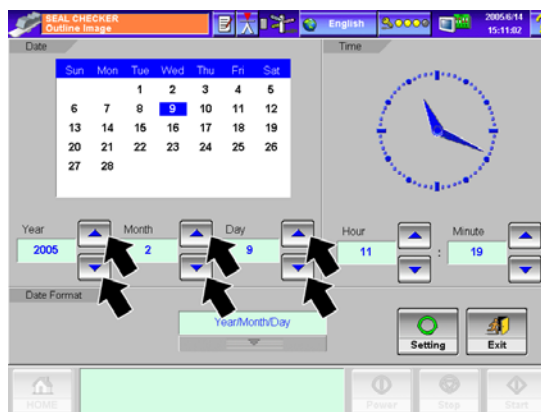


Fig.5-33 Time Setting Screen

3. Enter the time with the Up and Down keys  or the ten-key pad.

NOTE

Time cannot be entered directly from the clock face.

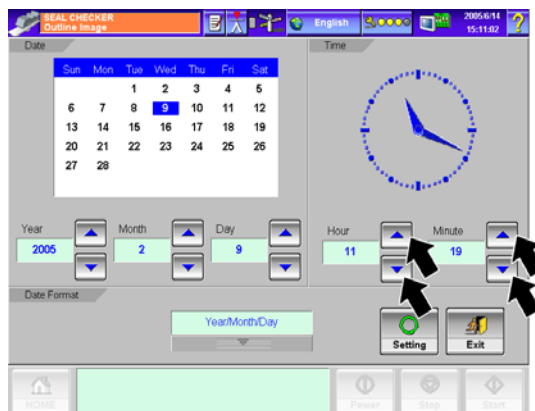


Fig.5-34 Time Setting Screen

4. Press the drop-down list key  to select year, month, and date.

5. Press the Setting key .
- The clock will start with the set date and time.

NOTE

Second is 0 at the setting.

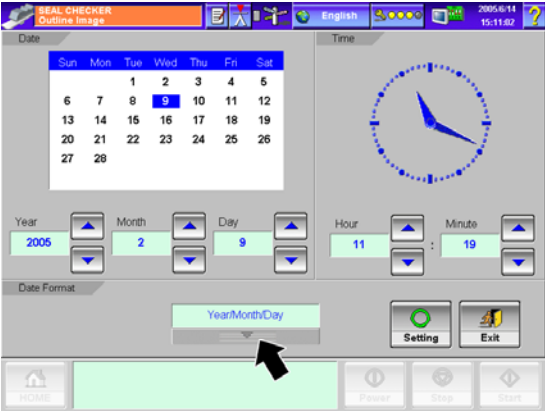


Fig.5-35 Time Setting Screen

6. Press the Exit key  to end the Time Settings screen.

5.3.6 Help function

This function gives you brief explanation about keys on the screen.

1. Press the Help key  on a screen.
 ► Help key blinks.



Fig.5-36 Main Menu Screen

2. Press the key which you want to know. (In this case, " SETTING" is pressed.)



Fig.5-37 Main Menu Screen (Help key blinks)

- The functional description of the pressed key appears.
 3. The Help display disappears when you touch any place of the screen.
 ► Blinking of the help key is released and the screen back to the normal screen.

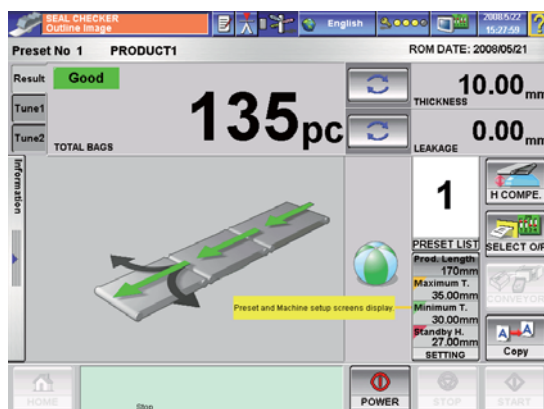


Fig.5-38 Help display Screen

5.4 MAIN MENU Screen

This section describes functions of the screens and instructions for operating them.

Switching of the screens or settings of the values can be performed by touching directly the keys displayed on the operation panel of the remote control unit.

The Main Menu screen appears first when the equipment power is turned ON. To start and stop operation or modify the equipment settings, call up the relevant menu from this screen.

CAUTION

- Use finger to press the display keys on the operation panel.
Using a ball point pen or other pointed object can damage the operation panel.
- The dimmed key is not in effect. This is the function for the other operational level or an optional function.
Applying excessive force to the operation panel may result in damage.
Check the operational levels referring to Section 5.3.3.

NOTE

- For the basic operation procedures, refer to " Chapter 6 ".

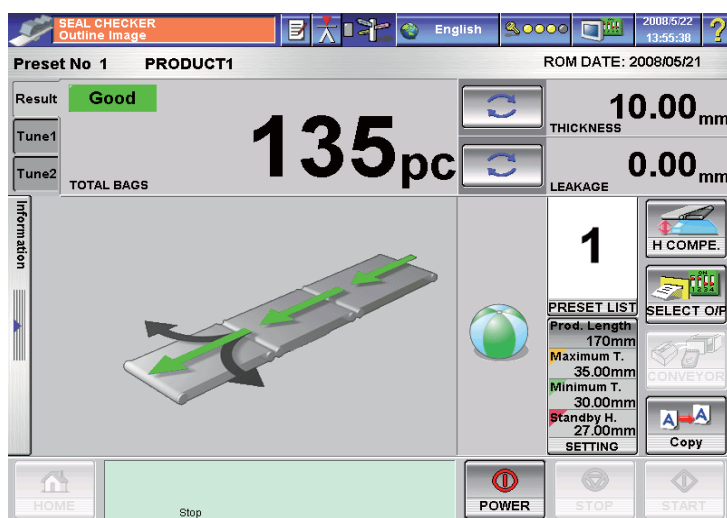


Fig.5-39 Main Menu Screen

NOTE

- The above figure shows the Main Menu screen after the operation level is changed to " Installation engineer level ". Normally, the screen is invoked with the " Operator level " .

Key or Tab	Function
Result tab	Displays the Result screen on Production screen and Main Menu screen. Refer to 5.4.2.1
Tune1 tab	Displays the Tune1 screen on Production screen and Main Menu screen. Refer to 5.4.2.2
Tune2 tab	Displays the Tune2 screen on Production screen and Main Menu screen. Refer to 5.4.2.3
Information	Displays the INFORMATION screen for statistics, error log, etc. Refer to 5.10
PRESET LIST key	Displays the PRESET LIST screen. Refer to 5.5
SETTING key	Displays the Preset Parameter screen. The major set values are displayed above the key. Refer to 5.8
H COMPE. key	Executes Height Compensation. (Refer to “8.3.4” Height Compensation of Seal Check Conveyor”)
SELECT O/P key	Displays the Output Select screen. (Refer to “5.7 Output Selection”)
CONVEYOR key	Discharges products on the conveyor. This function is valid when the power is ON and the system is stopped.
COPY key	Displays the Copy screen. (Refer to “5.6 Preset Copy”)
POWER key	Turns ON and OFF the drive power of the system. When the power is ON, the key lights up. This function is invalid during operation.
STOP key	Stop the system.
START key	Starts the system.
HOME key	Displays the LINE CONTROL HOME MENU Screen. (applied with ITPS)

5.4.1 ZOOM CHANGEOVER

Zoom changeover function is the function in which one of weighed value, quantity, capacity displayed on the Main Menu or Operation Menu can be zoomed up on the display.

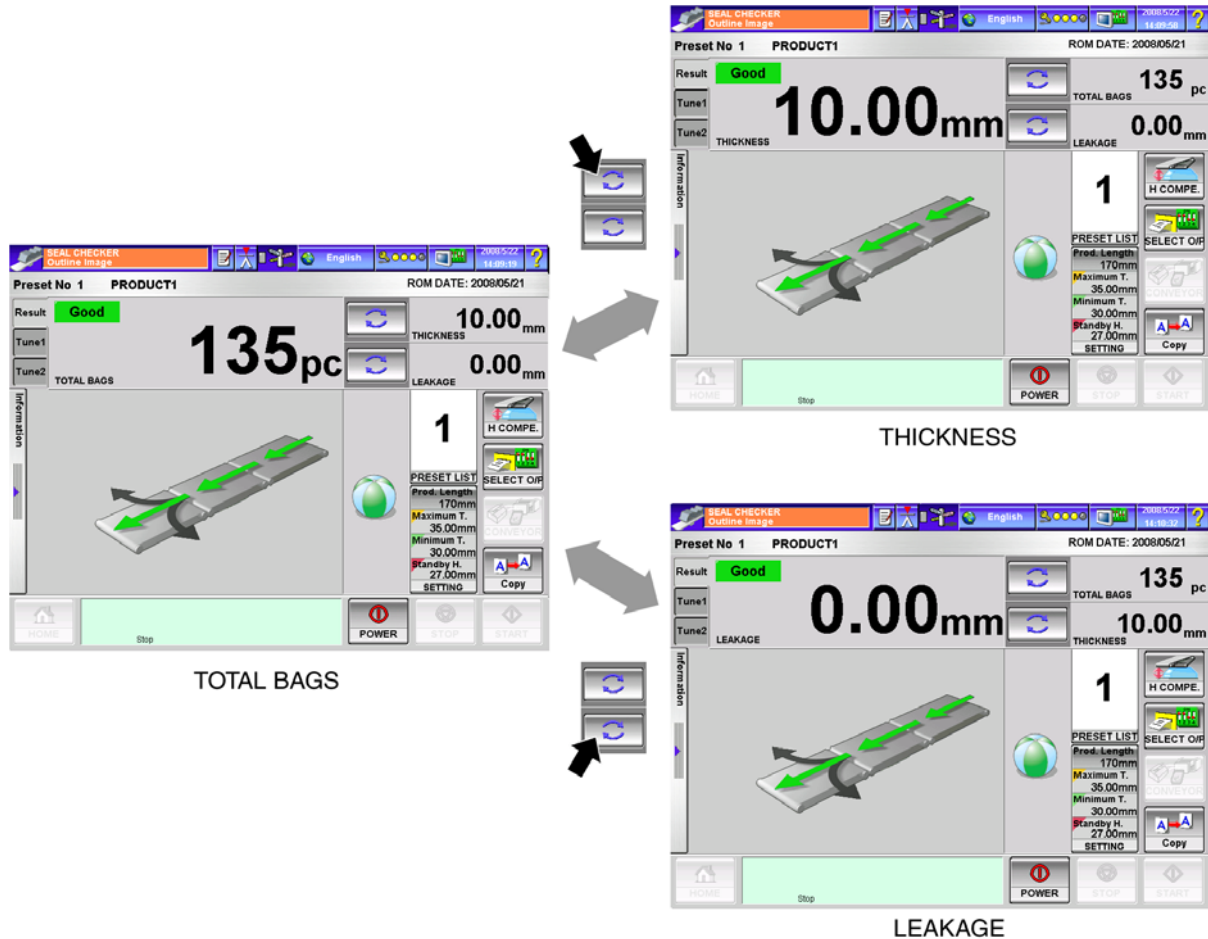


Fig.5-40 Zoom Changeover Screen

< Operation procedure >

The Zoom changeover keys  can be used to select one of the measured thickness, quantity and capacity in the zoom field.

Pressing the upper Zoom changeover key replaces the item on the right of the key with the item on the left of the key. Pressing the lower Zoom changeover key replaces the item on the right of the key with the item on the left of the key.

TIP

- The capacity display can be changed over between the average display and the standard deviation display.

5.4.2 Left Tab

Press any tab (Result, Tune1, or Tune2) on either the Main Menu screen or Outline Image screen (Production screen). The Result screen, the Tune1 screen, or Tune2 screen will appear.



Fig.5-41 Main Menu / Result Screen

Display	Function
Result	Refer to "5.4.2.1 Result".
Tune1	Refer to "5.4.2.2 Tune1".
Tune2	Refer to "5.4.2.3 Tune2".

5.4.2.1 Result

The Result function displays the detection result of product. The following values are displayed: the total product quantity, the measurement value of air leakage, and the measurement value of thickness

1. Press the Result tab on the screen.

►The Result screen appears.



Display	Function
TOTAL BAG	Total product quantity

Display	Function
THICKNESS	Measurement value of thickness
LEAKAGE	Measurement value of air leakage

5.4.2.2 Tune1

The Tune1 function adjusts the range of judged value.

1. Press the Tune1 tab on the screen.

►The Tune1 screen appears.



Display	Function
Standby Height	Refer to " 5.8.2.1 CRITERIA ".
Minimum Thickness	Refer to " 5.8.2.1 CRITERIA ".
Maximum Thickness	Refer to " 5.8.2.1 CRITERIA ".
Maximum Leakage	Refer to " 5.8.2.1 CRITERIA ".

5.4.2.3 Tune2

The Tune2 function adjusts the speed of each conveyor.

1. Press the Tune2 tab on the screen.

►The Tune2 screen appears.



Key or Display	Function
Reject Speed	Reject conveyor speed is set by the unit of 1m/min.

Key or Display	Function
Check Speed	Check conveyor speed is set by the unit of 1m/min.
Infeed Speed	Infeed conveyor speed is set by the unit of 1m/min.
J-Belt Speed	J-Belt conveyor speed is set by the unit of 1m/min.
 m/min	Press this key to display the ten-key. Enter and set the numerical value using the ten-key.
	A single pressing the button increases or decreases the setting by ± 1 . Hold down the key to automatically increase/decrease the value by ± 1 .

5.4.3 H COMPE.

The H COMPE. function can be operated in the Site Engineer level or upper levels. This function allows the test product to flow and checks the thickness. For operation procedures, refer to " 8.3.4 Height Compensation of Seal Check Conveyor ".

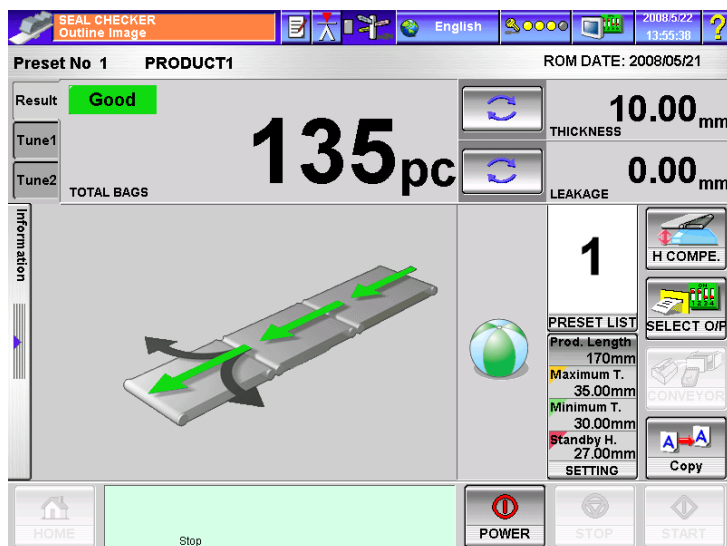


Fig.5-42 H COMPE. Screen

5.5 PRESET LIST Screen

5.5.1 Photo Display

Press the photo part of the preset display (PRESET LIST key)  on Main Menu screen to display the PRESET LIST screen.

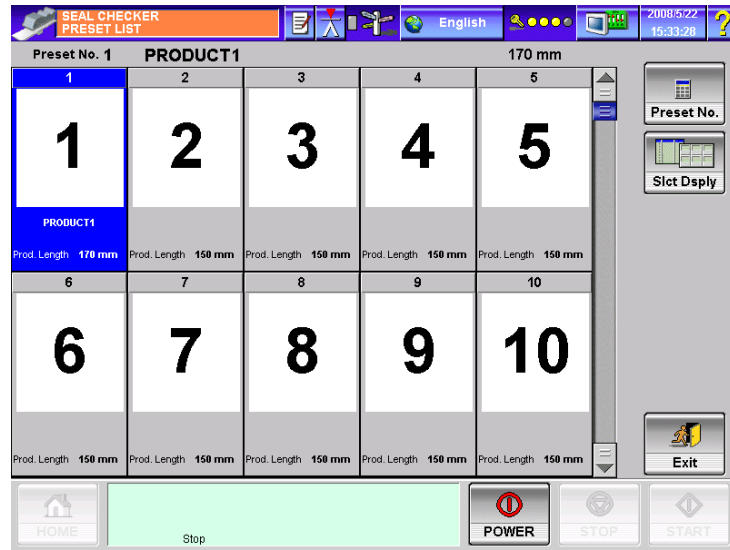


Fig.5-43 PRESET LIST Screen

Table 5-8 Setting Items and Functions on the PRESET LIST Screen (Photo display)

Key and Display	Function
Select preset display	Displays the selected the preset number, product name, and target thickness.
Select Preset key 	Each row is a key to open preset. Displays specification numbers, preset numbers, names of products, target length (or thickness), speed, number of discharging, and revised timing. The photo is displayed if there are any photo data.
Preset No.key	Selects the Preset No. directly via ten-key.
Slct Dsply key	Switches the display method to the list display method.
Exit key	Returns to the Main Menu screen.

5.5.2 List Display

Press the Slct Disply key  on the PRESET LIST screen to display the list screen.

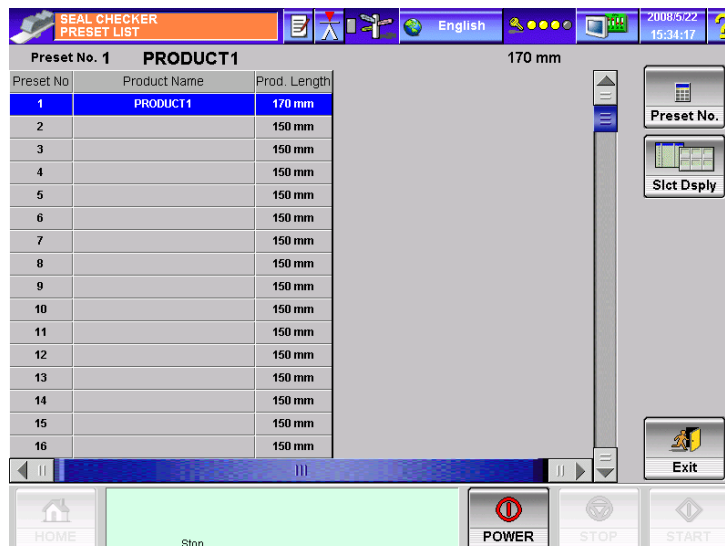
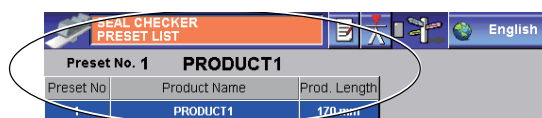


Fig.5-44 PRESET LIST Screen (List display)

Table 5-9 Setting Items and Functions on the PRESET LIST Screen (List display)

Key and Display	Function
Select preset display	Displays the selected the preset number, product name, and target thickness.
Select Preset key	Open the preset. The preset no., product name, target length are displayed on the key. The photo is displayed if there are any photo data.
Preset No. key	Selects the Preset No. directly via ten-key.
Slct Disply key	Switches the display method to the photo display method.
Exit key	Returns to the Main Menu screen.

Each option on the list can be sorted. When the header of each row (e.g. Product Name) is pressed, the items are rearranged in ascending sequence by using the pressed row as the basis.



5.6 PRESET Copy

The Preset Copy screen is called up by pressing the Copy key in the MAIN MENU screen. Performs copy or initialization of presetting.

NOTE

- Preset Copy is available at the installation level or over level.

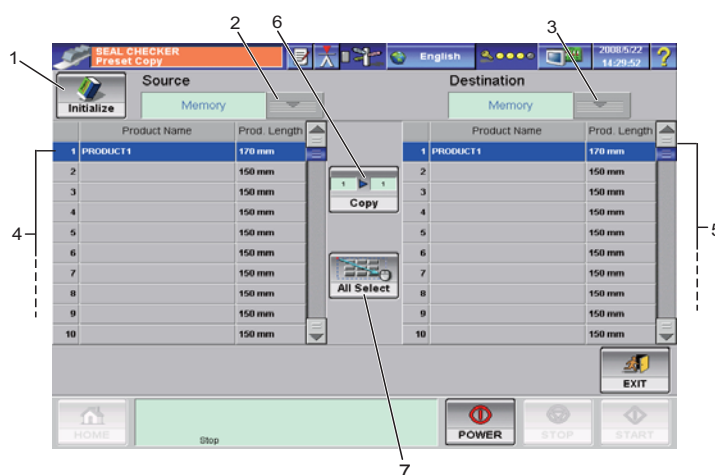


Fig.5-45 Preset Copy Screen

Table 5-10 Setting Items and Functions on the Preset Copy Screen

No.	Setting item	Function
1	Initialize	Initializes the preset contents of the selected preset number.
2	Copy source selection	Selects the agent of the copy source from "Memory" and "Card".
3	Copy destination selection	Selects the agent of the copy destination from "Memory" and "Card".
4	List of copy source preset	Selects the preset number of the copy source.
5	List of copy destination preset	Selects the preset number of the copy destination.
6	Copy	Execute preset copy from the copy source to the copy destination.
7	All Select	Selects all sources of copy.
8	Exit	Returns to the Main Menu screen.

5.6.1 Copying selected preset

This section explains the procedure to select and copy the preset registered in the memory or the card.

NOTE

- If the data has been already registered in the designated copy destination, the old data is overwritten by the new data by executing copy.

- Press the Copy Source dropdown key and select a desired copy source from "Memory" or "Card".

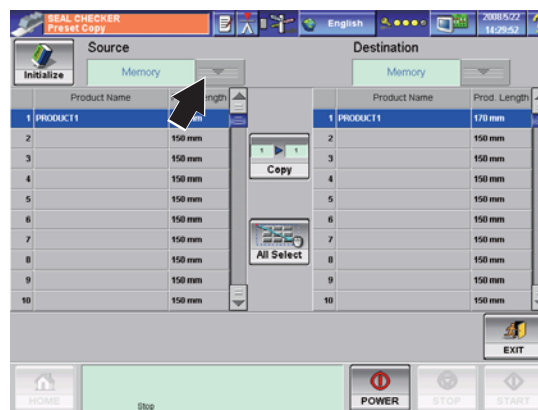


Fig.5-46 Preset Copy Screen

- Press the preset number to be the copy source from the list of copy source preset.

► The selected preset source is displayed in blue.

► The number of preset copy source is

displayed on the left of the Copy key

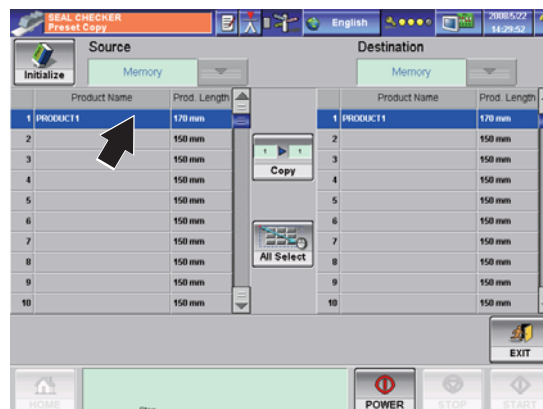


Fig.5-47 Preset Copy Screen

3. Press the Copy Destination dropdown key and select a desired copy destination from "Memory" or "Card".

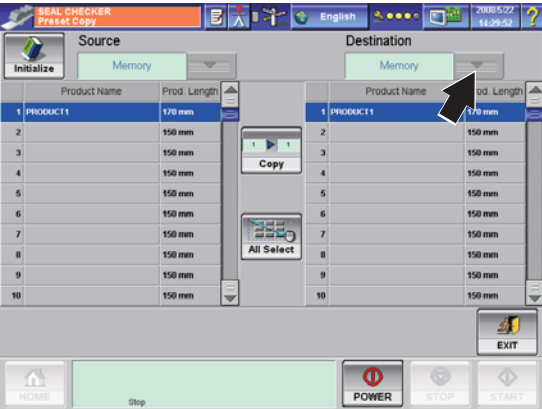


Fig.5-48 Preset Copy Screen

4. Press the preset number of the copy destination from the list of copy destination preset.

- ▶ The selected preset item of the copy destination is displayed in blue.
- ▶ The preset number of the selected copy destination is displayed on the right of the

Copy key .

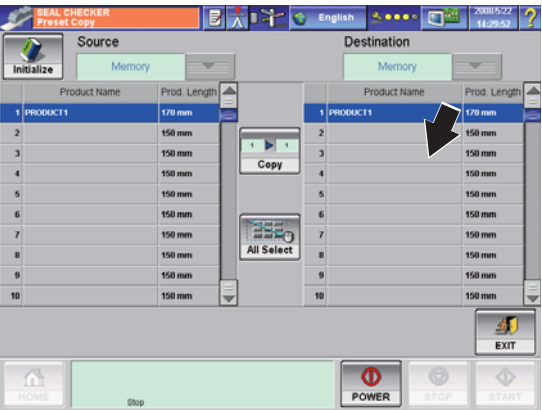


Fig.5-49 Preset Copy Screen

5. Press the Copy key .

- ▶ The message "Wait for a moment" appears and copy is executed.

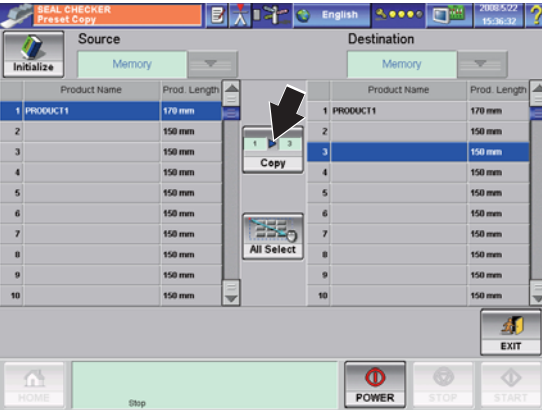


Fig.5-50 Preset Copy Screen

- ▶ The copied preset is displayed on the selected copy destination.
- ▶ Preset copy selection is complete.

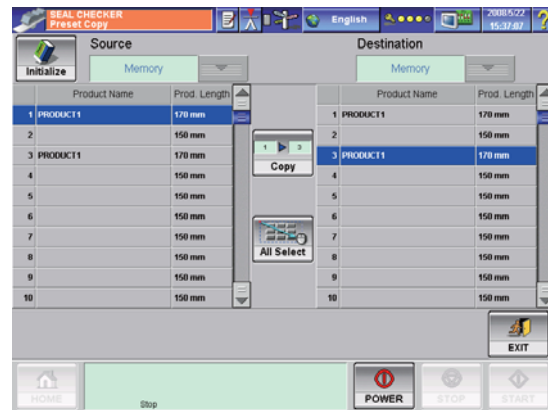


Fig.5-51 Preset Copy Screen

5.6.2 Copy all

This section explains the procedure to copy all preset registered in the memory or the card. When copying all selection, the copy is from the memory to the card, or the card to the memory.

NOTE

- Once all selection copy is executed, the old data is overwritten by the new data and the old data is lost.

1. Press the Copy Source dropdown key and select a desired copy source from "Memory" or "Card".

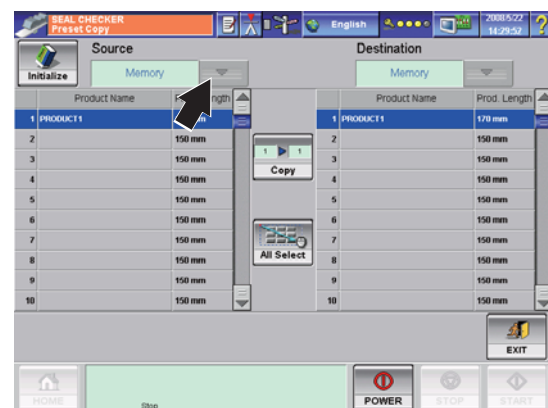


Fig.5-52 Preset Copy Screen

2. Press the All Select key .
- All preset items are displayed in blue.

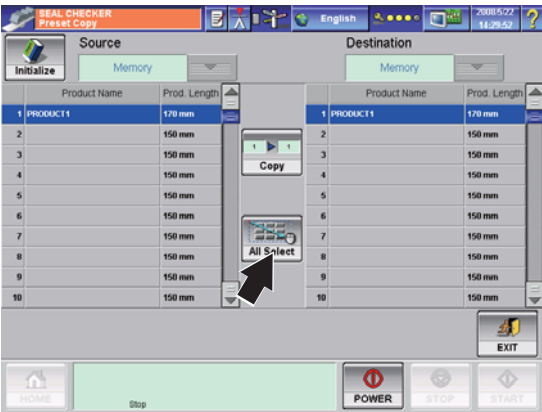


Fig.5-53 Preset Copy Screen

3. Press the Copy Destination dropdown key and select a desired copy destination from "Memory" or "Card".

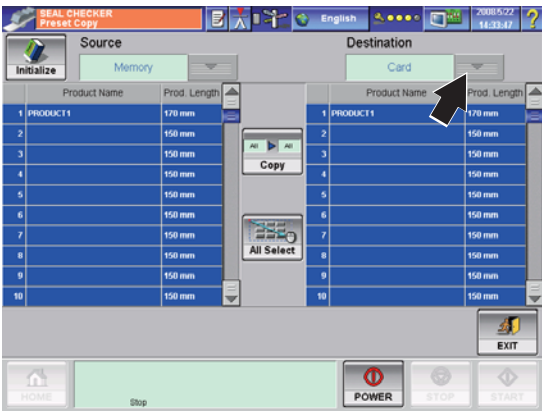


Fig.5-54 Preset Copy Screen

4. Press the Copy key .
- The message "Wait for a moment" appears and copy is executed.

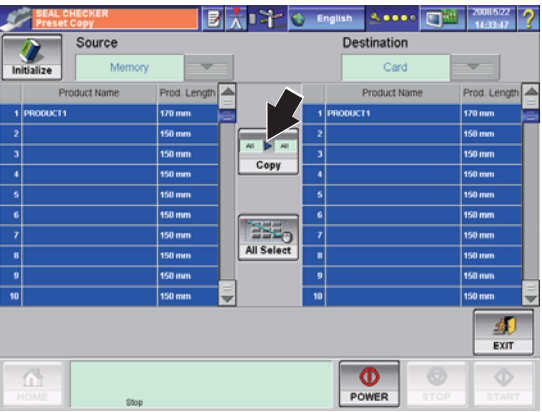


Fig.5-55 Preset Copy Screen

- All selection preset copy terminates.

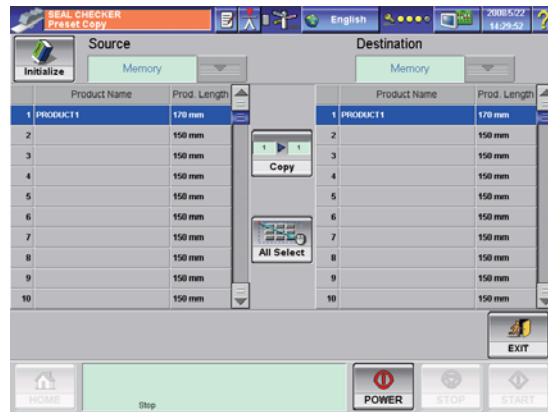


Fig.5-56 Preset Copy Screen

5.6.3 Selection and Initialization of Preset

This section explains the procedure to initialize a certain preset data.
For initialization of the preset, use the list of the preset copy source.

1. Press the Copy Source dropdown key and select a desired copy source from "Memory" or "Card".



Fig.5-57 Preset Copy Screen

2. Press the preset number for initialization from the list of the preset copy source.

- The selected preset item of the copy source is displayed in blue.

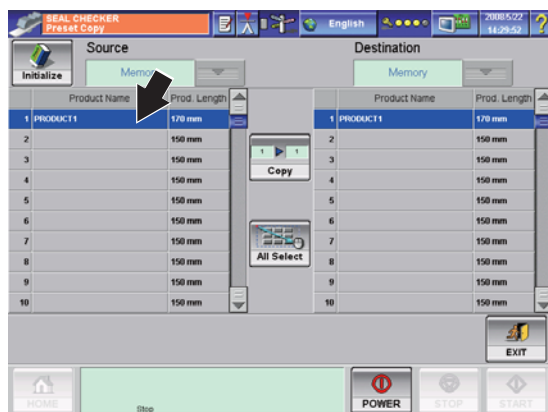


Fig.5-58 Preset Copy Screen

3. Press the Initialize key .

► The message "Wait for a moment" appears and initialization is executed.



Fig.5-59 Preset Copy Screen

5.6.4 Initialization of all

This section explains the procedure to initialize the all preset data.
In the all preset initialization, use the list of the preset copy source.

1. Press the Copy Source dropdown key and select a desired copy source from "Memory" or "Card".



Fig.5-60 Preset Copy Screen

2. Press the All Select key .

► All preset items are displayed in blue.



Fig.5-61 Preset Copy Screen

3. Press the Initialize key .

►The message "Wait for a moment" appears and initialization is executed.

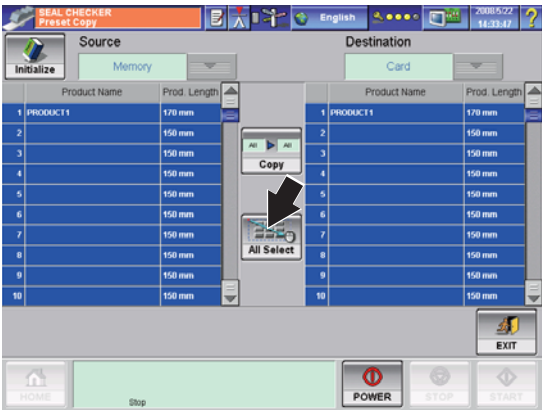


Fig.5-62 Preset Copy Screen

►All selection preset initialization is complete.

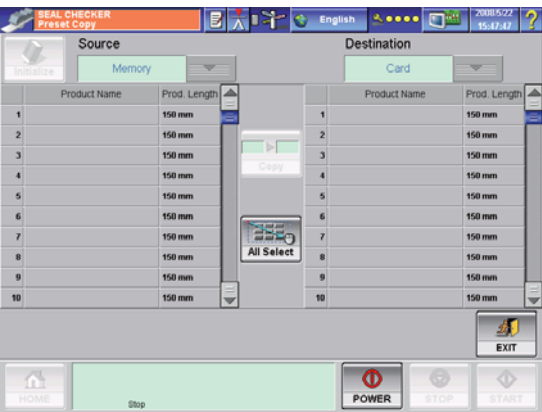


Fig.5-63 Preset Copy Screen

5.7 Output selection

The output selection screen is called up by pressing the SELECT O/P key on the Main manu screen. The output selection screen allows the user to select an item and outputs the data to a printer or a compact flash card. The selected item data are outputted to the output destination specified in the Control panel. To output a data to the printer, the printer (option) must be connected to the equipment.

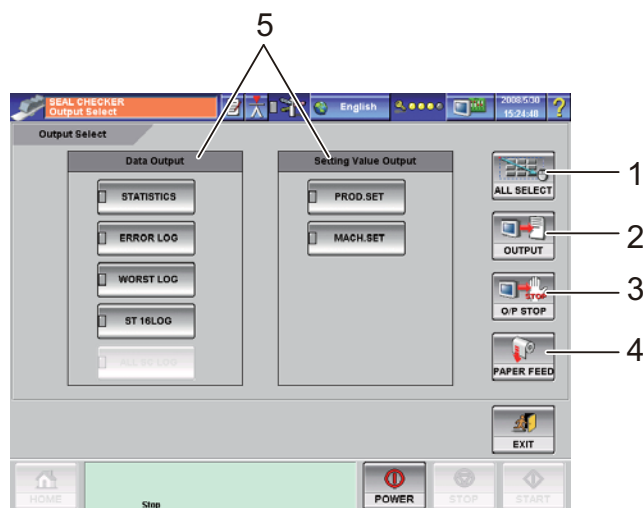


Fig.5-64 Output Selection Screen

No.	Name	Symbol	Function
1	All select key		Selects all items.
2	Output key		Outputs the selected item data.
3	Output stop key		Stops the data output.
4	Paper feed key (only when a printer is connected)		Feeds paper by one line. Use this key to replace printer paper or set a blank space after each total line.
5	Information output/set value output key	—	Selects an item to be outputted. [ALL SC LOG] key cannot be simultaneously selected with the other item.

NOTE

- For the procedure to set the output destination, refer to "5.3.4.3 Data output destination" in Control panel.
- To output data to a compact flash card, it must be inserted in the compact flash card slot of the equipment.

NOTE**Selection of Output Items**

- All SC Log cannot be simultaneously selected with the other items.
- When the data output destination is set to “Printer”, ALL SC Log cannot be selected, and [PAPER FEED] key



is not displayed.

- Pressing the All Select key selects all the items except ALL SC Log.
-

5.8 PRESET PARAMETER

The PRESET PARAMETER screen is called up by pressing the SETTING(Preset contents display) key in the Main Menu screen.

A preset is a group of parameters necessary for processing a given product item. It eliminates the need to set parameters individually when the product is processed again.

To change the preset number, use the [▲] or [▼] keys or press the Preset number key to call up the ten-key pad and enter your desired preset number directly.

The PRESET PARAMETER screen contains the following three pages, Product Setting, Basic Setting, Advanced Setting.

For further details on each page, refer to the following paragraphs.

NOTE

- The Site Engineer level enables to set the items of Product Setting and Basic Setting.
- The Installation level enables to set the items of Product Setting, Basic Setting, and Advanced Setting.

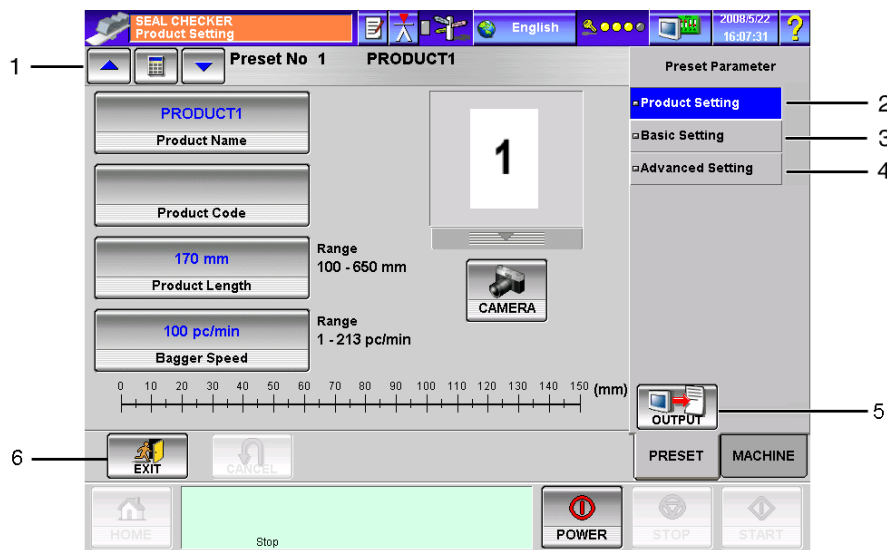


Fig.5-65 Preset Parameter Screen

No.	Key or Display	Function
1		Changes the preset number.
2	Product Setting	Displays Product Setting screen.
3	Basic Setting	Displays Basic Setting screen.
4	Advanced Setting	Displays Advanced Setting screen.
5	OUTPUT	Output the selected preset data.
6	EXIT	Settings are completed and returns to Main Menu screen.

5.8.1 Product Setting Screen

The Product Setting screen appears when calling up the Preset Parameter screen.

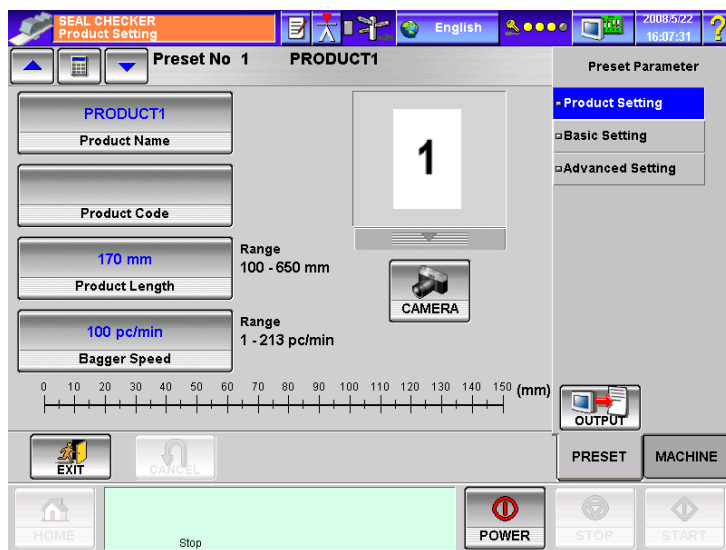


Fig.5-66 Product Setting Screen

Key or Display	Function
Product Name	Enter the product name using the keyboard. (Max. 24 characters)
Product Code	Enter the product code using the keyboard. (Max. 16 characters)
Product Length	Input Product length. Depending upon length entered, TSC-R automatically selects inspection mode; normal or pose. *1 Upon entering product length, TSC-R automatically calculates the conveyor speed and other timings based on the bagger speed. *2
Bagger Speed	Enter current bagger speed.
Photo select key 	Select the photo or illustration of a product.
CAMERA	Take photos of product.

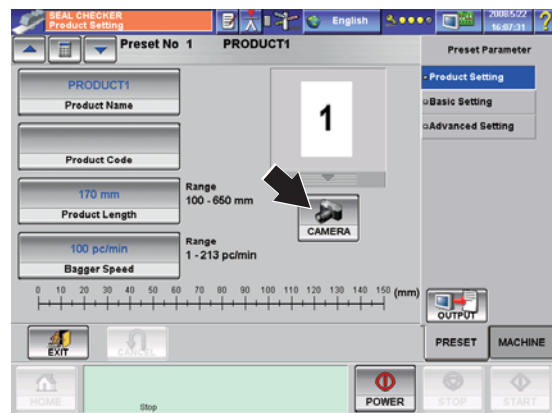
NOTE

- To automatically set the appropriate conveyor speed of product, configure the "Product Length" setting and the "Bagger Speed" setting (in this order).
First set these two items for the initial product setting.
- *1: Product length $\leq 248\text{mm}$ $\Rightarrow$ Normal mode inspection will be selected automatically.
Product length $> 248\text{mm}$ $\Rightarrow$ Pose mode inspection will be selected automatically.
This value changes depending on the standby height.
If you wish to use pose mode inspection, please set the unit with following procedure:
- Set product length.
 - Manually change the inspection mode to pose mode
(PRESET>Basic Mode/Speed Setting)
 - Set Bagger Speed.
- *2: Upon changing the product length, and if the bagger speed set in RCU on the product length exceeds the theoretical bagger speed at that length. In this case, a confirmation on bagger speed will be displayed on RCU screen, therefore please adjust the bagger speed based on the indication. (Refer "5.8.2.4 Timing Auto Calculation".)

5.8.1.1 Taking photos of product

- Press the Camera key .

► A message to confirm the photo taking will appear.

**Fig.5-67 Product Tab Menu**

- Point the product at the camera, and press the Yes key .

► The taken photo will be displayed.
The image taken by the camera is displayed on the left, and the image previously taken is displayed on the right.

**Fig.5-68 Message to Confirm the Photo Taking**

3. Press the Exit key  when not taking a photo. Press the Photograph key  when taking a photo.

► When Exit key is pressed, the previous menu will appear without taking any photo.
When pressing the Photograph key, the taken image will be displayed on the right.



Fig.5-69 Taken Photo

4. Press the Cancel key  when retaking another photo. Press the Save key  when saving the photo.

► The taken photo will be saved.



Fig.5-70 Taken Photo

5. Press the Exit key .

► It will return to the previous menu.

► The taken image will be registered in the photo list.



Fig.5-71 Taken Photo

NOTE

- Images are not shown when the camera is not connected.
- Enough light needs to be focused on product when taking photos. The images taken in dark surroundings can be inferior in quality.
- Only one photo data can be registered for each preset.
A new photo is replaced with the old photo when taking a picture with the already registered preset number.

5.8.2 Basic Setting Screen

The Basic Setting screen is called up by pressing the Basic Setting index on the Preset Parameter screen.

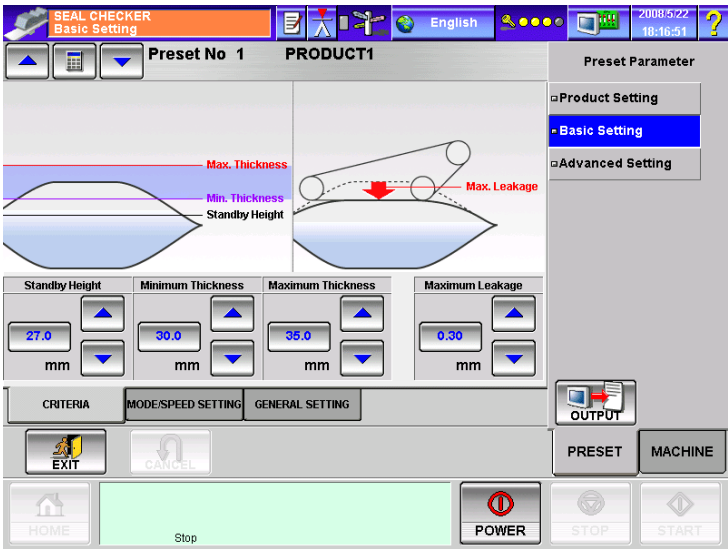


Fig.5-72 Basic Setting Screen

Key or tab	Function
CRITERIA	This sets the range of judged value. Refer to " 5.8.2.1 CRITERIA"
MODE/SPEED SETTING	This sets the speed of each conveyor. Refer to " 5.8.2.2 MODE/SPEED SETTING"
GENERAL SETTING	This sets the output timing to the external equipment. Refer to " 5.8.2.3 GENERAL SETTING "

5.8.2.1 CRITERIA

The CRITERIA screen is called up by pressing the CRITERIA tab on the Basic Setting screen. The CRITERIA screen is used for setting the critical function of seal checker.

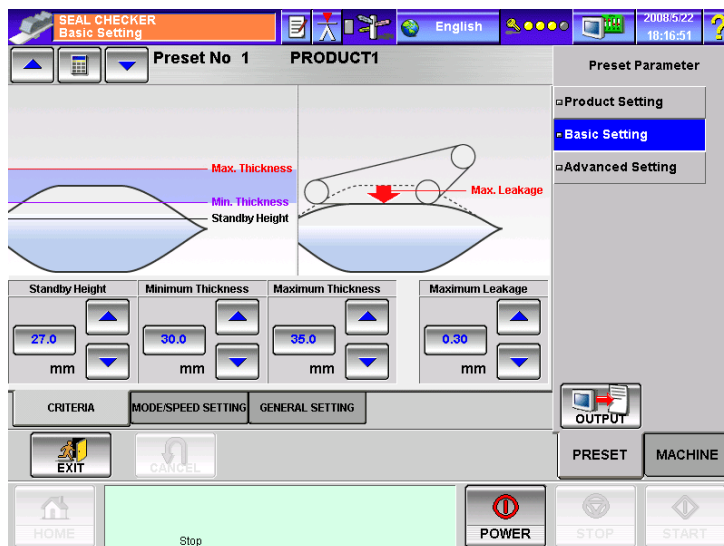


Fig.5-73 CRITERIA Screen

Key or Display	Function
Standby Height	The stand-by height of conveyor above the seal check section is set. (Ref. Note) Set the value of approx. 5mm smaller than the bag thickness.
Minimum Thickness	If the thickness of the inspected bag is smaller than this value, the bag is determined to be a "flat bag". (Ref. Note)
Maximum Thickness	If the thickness of the inspected bag is greater than this value, the bag is determined to be a "thick bag". (Ref. Note)
Maximum Leakage	If the height change value exceeds this setting value, the bag is determined to be an "air leak" bag.
 mm	Press this key to display the ten-key. Enter and set the numerical value using the ten-key.
 	A single pressing the button increases or decreases the setting by ± 1 . Hold down the key to automatically increase/decrease the value by ± 0.1 or ± 0.01 respectively for "Minimum or Maximum Thickness" or "Maximum Leakage".
EXIT	Exits from the Product Parameter screen.

(Note) Refer the following procedure about the setting item 2 to 4.

1. Put some bags to check (good bags) and measure the bag thickness.

For example, the bag thickness measured approx. 30mm $\pm$ 2mm of range.

Set [Maximum Thickness] to 35mm

Set this value per local regulations. For example the thickness can be allowed to +5mm (35mm).

Set [Minimum Thickness] to 27mm

Set this value to -1mm of the thinnest bag among the measured bags.

Set [Standby Height] to 25mm

Set this value to -5mm of average thickness.

NOTE**Points of setting**

- Make sure to change enough air to products.
- Always set values so that they become [Maximum Thickness] > [Minimum Thickness] > [Standby Height].

5.8.2.2 MODE/SPEED SETTING

The MODE/SPEED SETTING screen is called up by pressing the MODE/SPEED SETTING tab on the Basic Setting screen.

The MODE/SPEED SETTING is used to set the conveyor speed of Check conveyor, Infeed conveyor, and J-belt conveyor. This screen is also used to set the speed when the optional bag stabilizer is connected.

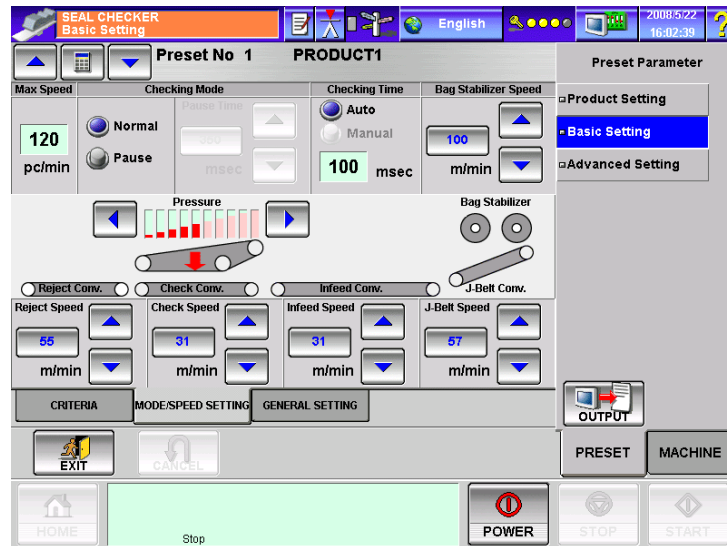


Fig.5-74 MODE/SPEED Screen

Key or Display	Function
Max Speed	When the speed or inspection condition of each conveyor is set, the Max Speed is displayed after the auto calculation of the highest theory capability.
Checking Mode	Normal: Performs continuous operation. Pause: Temporarily stops at the check conveyor to measure.
	Pause Time: Sets the measurement time when "Pause Mode" is set to "Pause".
Checking Time	Auto: Timing parameters are automatically calculated. Manual: The user can customize timing parameters.
Bag Stabilizer Speed	Sets the rotation speed of optional bag stabilizer.

Key or Display	Function
Pressure	Adjusts the pressure to the products. Press   to adjust the level. This links to "No.1" of "Advanced Setting screen". Refer to " 5.8.3 Advanced Setting " When "Pause Mode" is set to "Pause", the pressure is "Pause Mode Pressure". This links to "No.3" of "Advanced Setting screen". Refer to " 5.8.3 Advanced Setting "
Reject Speed	Set the Reject conveyor speed. (1 m/min)
Check Speed	Set the Seal check conveyor speed. (1 m/min)
Infeed Speed	Set the Infeed conveyor speed. (1 m/min)
J-Belt Speed	Set the J-belt conveyor speed. (1 m/min)
 m/min	Press this to display ten-key. Enter a numerical value using the ten-key to set the conveyor speed.
 	A single pressing the button increases or decreases the setting by ± 1 . Hold down the key to automatically increase/decrease the value by ± 1 .
EXIT	Exit from the Product Parameter screen.

5.8.2.3 GENERAL SETTING

The GENERAL SETTING screen is called up by pressing the GENERAL SETTING tab on the Basic Setting screen.

The GENERAL SETTING screen is used to set the input/output time to the external equipment on this screen.

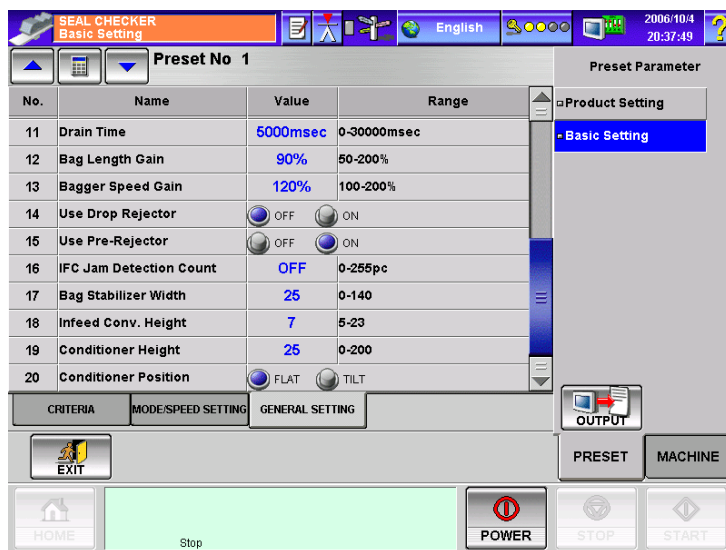


Fig.5-75 GENERAL Screen

Key or Display	Function
Continuous N/G Count	If the frequency of continuous N/G bags exceeds the set value for "continuous N/G quantity", all subsequent bags are determined to be "continuous N/G". When "continuous N/G reject direction" is set to "Stop" in the reject setting, this system stops at occurrence of an error. When set to "0," the system does not stop even if N/G bags are continuously detected.

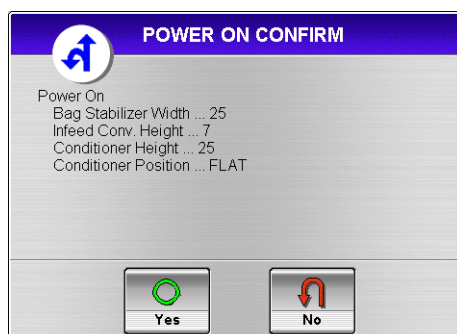
Key or Display	Function
Reject Delay Time	Set the time after inspection until the reject device starts moving. *1
Reject ON Time	Set the time the reject device operates. *2
Output Sig.1 Delay Time	After inspection is completed, output signal 1 turns ON after the lapse of this time.
Output Sig.2 Delay Time	After inspection is completed, output signal 2 turns ON after the lapse of this time.
Output Sig.3 Delay Time	After inspection is completed, output signal 3 turns ON after the lapse of this time.
Output Sig.4 Delay Time	After inspection is completed, output signal 4 turns ON after the lapse of this time.
Ext. Input 1 Delay Time	The time to check the external N/G 1 input signal is set going back from the detection time of the tip of a bag.
Ext. Input 2 Delay Time	The time to check the external N/G 2 input signal is set going back from the detection time of the tip of a bag.
Stop Delay Time	After pressing the STOP key, conveyor stops after the lapse of this time.
Bag Length Gain	Product length ration set in ITPS screen and actual length on TSC-R. Ex): Bag Length Gain: 90%, and the length in ITPS screen is set 200mm, 180mm (90% of ITPS setting) will be set on TSC-R screen. This feature is applied for ITPS interface only.
Bagger Speed Gain	Ration of bagger speed and theoretical speed. Ex): Bagger speed gain: 120% and bagger speed 100bpm is set, theoretical max. speed will be automatically calculated to 120bpm (more than 120% of bagger speed of 100bpm).
Use Drop Rejector	This parameter will decide the reject method (when “air + drop-belt” or “air + 2-step drop belt” is used). OFF: Air 3-way reject will be used. ON: 2-way Drop-belt will be used. *3
Use Pre-Rejector	Select ON/OFF of pre-rejector. OFF: Pre-rejector will not operate. ON: Pre-rejector will operate.
IFC Jam Detection Count	Set the numbers of bags to detect jamming on incline conveyor. 1-255: Conveyor stops by error when successive missing bags of set numbers are detected. 0: Jam detection will not be used.
Bag Stabilizer Width	Adjust the bag stabilizer width (unit of mm). *4 Usually enter bag thickness (unit of mm) . If automated adjustment unit is equipped, bag stabilizer move to the setting location. *5
Infeed Conv. Height	Adjust the infeed conveyor height (unit of inches). *4 Usually enter bag length (unit of inches). If automated adjustment unit is equipped, infeed conveyor move to the setting location.
Conditioner Height	Set the height of conditioning brushes. *4 Usually enter bag thickness.
Conditioner Position	Set the angle of conditioning brushes. *4 Depending upon product condition, choose flat / tilt.
IFC. MAX. SPEED in AutoCal	Set the maximum speed of the infeed conveyor in auto calculation mode. *6
IFC. MIN> SPEED in AutoCal	Set the minimum speed of the infeed conveyor in auto calculation mode. *6

*1: This delay time is only for air reject and this is set automatically. This parameter will not be used for drop-down reject (note: reject delay time for drop-belt reject is 0). However, when air and 2-step drop-belt are set as reject method and 2-step drop belt is used, this parameter controls reject delay time of 2-step drop-belt. A manual setting is required in this case.

*2: This is the ON time for air jet reject, and the air nozzle blows per this time. This time is calculated automatically.
This ON time will not be used for drop-belt reject.

*3: In drop-belt reject, all improper items will be rejected by drop-belt; the reject direction setting (right or left) will not be used. After rejecting improper item, the drop-belt stays the same position until next correct item, which goes to centerline, comes.

*4: These parameters will be confirmed as per confirmation screen below before starting operation. Please double-check if the parameters are set correctly prior to operation.



*5: These parameters will be adjusted automatically when power is on. If these parameters are changed during operation, adjustment will be done automatically. Please be careful not to catch your hands to the moving parts. (Refer “5.12 Automatic Adjustment Unit”)

*6: Conveyor speed is automatically calculated.

Timing auto calculation will be done in setting “IFC.MAX.SPEED in AutoCal” and “IFC.MIN. SPEED in AutoCal”. (Refer “5.8.2.4 Timing Auto Calculation”).

5.8.2.4 Timing Auto Calculation

When the following parameters are changed on TSC-R, conveyor speed and inspection timings will be automatically calculated.

1. Product Length
2. Bagger Speed
3. Standby Height
4. Checking mode
5. Pause Time
6. Conv. Speed

7. Bagger Speed Gain

8. IFC.MAX.SPEED in AutoCal

9 IFC.MIN.SPEED in AutoCal

When these parameters are changed, bagger speed would be changed as well based on automatic calculation. In this case, there will be a confirmation dialog in RCU screen, please check the setting of bagger speed and adjust the setting if any change required. (Refer to “9.4 Warning Messages”.)

5.8.3 Advanced Setting

The Advanced Setting screen is called up by pressing the Advanced Setting index on the Preset Parameter screen.

The Advanced Setting screen is used for settings related to the pre-rejector if holding pressure is adjusted.



Fig.5-76 Advanced Setting Screen

Key or Display	Function
Assist Level	Adjusts the pressure to the products in the Normal Mode.
Prssure Limitation	The maximum value of Assist level that can be set in the Basic Setting.
Pause Mode Torque	The Assist Level in the Pause mode.
Pause Mode Filter Time	The inspection filter time in the Pause mode.
Damper Slow-Up Gain	Damper coefficient is set for the upper seal check conveyor to go up at slow speed. A greater value makes the damper stronger and a smaller value makes it weaker.
Damper Slow-Down Gain	Damper coefficient is set for the upper seal check conveyor to go down at slow speed. A greater value makes the damper stronger and a smaller value makes it weaker.
Damper Fast-Up Gain	Damper coefficient is set for the upper seal check conveyor to go up at high speed. A greater value makes the damper stronger and a smaller value makes it weaker.

Key or Display	Function
Damper Fast-Down Gain	Damper coefficient is set for the upper seal check conveyor to go down at high speed. A greater value makes the damper stronger and a smaller value makes it weaker.
Damper Up threshold	A threshold value for switching the low speed UP damper gain and high speed UP damper gain.
Damper Down threshold	A threshold value for switching the low speed UP damper gain and high speed UP damper gain
Short Judge Time	If the shading time by the product passing through the photoelectric sensor is shorter than this time, it is determined to be N/G and rejected.
Long Judge Time	If the shading time by the product passing through the photoelectric sensor is longer than this time, it is determined to be N/G and rejected.
Pre-Reject Delay Time	After the lapse of this time from the time when the tip of product is detected, reject operation is started.
Pre-Reject ON Time	After the lapse of this time from the start of reject operation, reject operation is finished. If this value is "0", reject operation is continued until a direct advance bag reaches.
Window after No-Good	All the products detected within this time after the rear end of a rejected product passed are determined to be N/G and reject.
Window before No-Good	If an acceptable product is contained within this time after the tip of a rejected product passed, instantaneously when it is determined to be N/G, reject operation is started.
Pre-Reject Auto Taiming	If this setting is ON, pre-reject timing is automatically calculated from the set values for conveyor speed and product length. If this setting is OFF, pre-reject timing is not calculated automatically. It must be set manually.

5.9 MACHINE PARAMETER

The MACHINE PARAMETER screen is called up by pressing the MACHINE tab on the Preset Parameter screen.

1. Press the [SETTING] key on the MAIN MENU screen.

► The Preset Parameter screen appears.

2. Press the [MACHINE] tab on the Preset Parameter screen.

► The Machine Parameter screen appears.

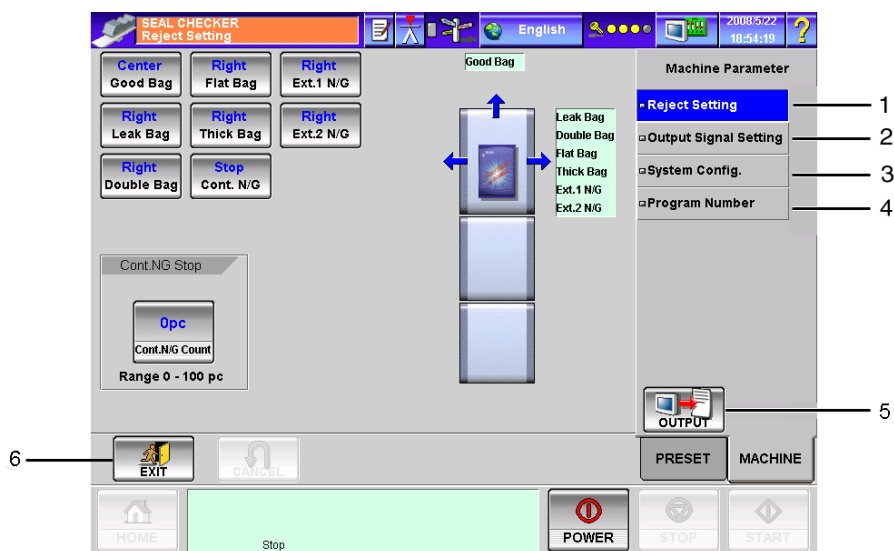
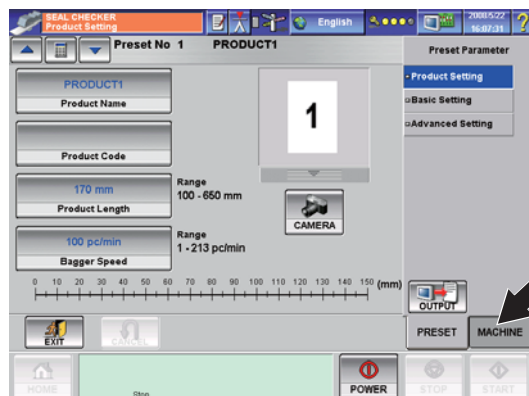


Fig.5-77 Machine Parameter Screen

No.	Key or Index	Function
1	Reject Setting	Changes to the Reject Setting screen. Refer to " 5.8.1 Reject Setting "
2	Output Signal Setting	Changes to the Output Signal Setting screen. Refer to " 5.8.2 Output Signal Setting "
3	System Setting	Changes to the System Setting screen. Refer to " 5.8.3 System Config. "
4	Program Number	Changes to the Program Number screen. Refer to " 5.8.4 Program Number "
5	OUTPUT	Outputs machine setting data.
6	EXIT	Settings are completed and returns to Main Menu screen.

5.9.1 Reject Setting

The Reject Setting screen first appears when calling up the Machine Parameter screen. The Reject Setting screen is used for seal checker reject direction setting.

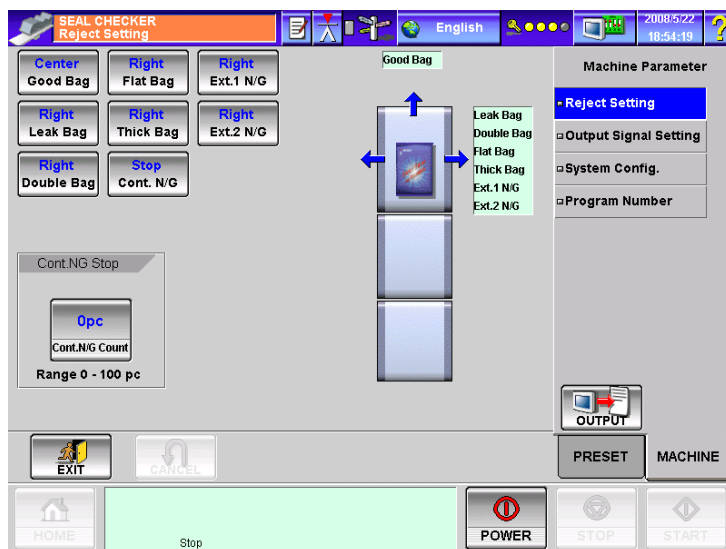


Fig.5-78 Reject Setting Screen

Key or Display	Function
Good Bag	Good bag reject direction is set. Select any one of left, center and right.
Flat Bag	Flat bag reject direction is set. Select any one of left, center, right and stop.
Ext.1 N/G	Ext.1 N/G reject direction is set. Select any one of left, center, right and stop.
Leak Bag	Air leak bag reject direction is set. Select any one of left, center, right and stop.
Thick Bag	Thick bag reject direction is set. Select any one of left, center, right and stop.
Ext.2 N/G	Ext.2 N/G reject direction is set. Select any one of left, center, right and stop.
Double Bag	Double bag reject direction is set. Select any one of left, center, right and stop.
Cont. N/G	Continuous N/G reject direction is set. Select any one of left, center, right and stop.
Cont. NG Count	Refer to " 5.8.2.3 GENERAL SETTING "No.1.

5.9.2 Output Signal Setting

The Output Signal Setting screen is called up by pressing the Output Signal Setting index on the Machine Parameter screen.



Fig.5-79 Output Signal Screen

Key or Display	Function
Good Bag Output Sig.	Good bag output signal is set.
Leak Bag Output Sig.	Leak bag output signal is set. *1
Flat Bag Output Sig.	Flat bag output signal is et. *1
Thick Bag Output Sig.	Thick bag output signal is set. *1
Double Bag Output Sig.	Double bag output signal is set. *1
Ext.1 N/G Output Sig.	Ext. 1 N/G output signal is set. *1
Ext.2 N/G Output Sig.	Ext.2 N/G output signal is set. *1
Cont. N/G Output Sig.	Continuous N/G output signal is set. *1
Output Sig.1 ON Time	Time to keep output signal 1 ON is set.
Output Sig.2 ON Time	Time to keep output signal 2 ON is set.
Output Sig.3 ON Time	Time to keep output signal 3 ON is set.
Output Sig.4 ON Time	Time to keep output signal 4 ON is set.

NOTE

- *1 For output signal setting, 4 types of output signals, namely "Output Signal 1", "Output Signal 2", "Output Signal 3" and "Output Signal 4", can be set in combination.

5.9.3 System Config.

The System Config. screen is called up by pressing the System Config. index on the Machine Parameter screen.

The existence of each conveyor and the type of reject device are set.

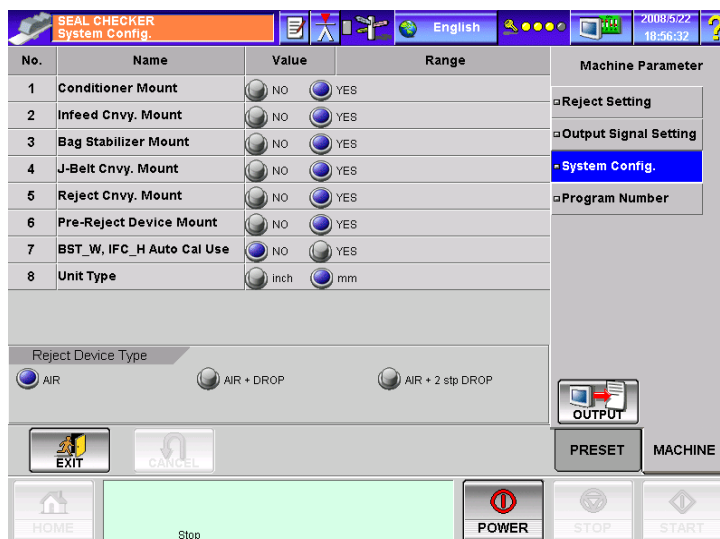


Fig.5-80 System Config. Screen

Key or Display	Function
Conditioner Mount	It is set whether the conditioning conveyor is equipped on the system. If this settings is YES, speed error detecting function, motor overload detecting function and speed correcting function are operative. If the setting is NO, these functions are inoperative.
Infeed Cnv. Mount	It is set whether the infeed conveyor is equipped on the system. If this settings is YES, speed error detecting function, motor overload detecting function and speed correcting function are operative. If the setting is NO, these functions are inoperative.
Bag Stabilizer Mount	It is set whether the bag stabilizer is equipped on the system. (Option) If this settings is YES, speed error detecting function, motor overload detecting function and speed correcting function are operative. If the setting is NO, these functions are inoperative.
J-Belt Cnv. Mount	It is set whether the J-belt conveyor is equipped on the system. If this settings is YES, speed error detecting function, motor overload detecting function and speed correcting function are operative. If the setting is NO, these functions are inoperative.
Reject Cnv. Mount	It is set whether the reject conveyor is equipped on the system. If this settings is YES, speed error detecting function, motor overload detecting function and speed correcting function are operative. If the setting is NO, these functions are inoperative.
Pre-Reject Device Mount	Select if pre-rejector is equipped or not. *1 NO: Select this when pre-rejector is not equipped. If this is selected, pre-rejector setting will be grayed out in Advanced Setting. YES: Select this when pre-rejector is equipped. If this is selected, pre-rejector setting will be available in Advanced Setting.

Key or Display	Function
BST_W, IFC_H Auto Cal use	No: The user has to adjust bag stabilizer width and infeed conveyor height. Yes: Bag stabilizer width and infeed conveyor height are automatically calculated.
Unit Type	Select the unit for product length setting. inch: Set the product length in inches. mm: Set the product length in mm.

*1: This is to gray out the unnecessary parameter for the unit. Please set the parameter referring actual machine condition.

Reject Device Type

Reject Device Type	Detail
AIR	Select when air 3-way reject is used.
AIR + DROP	Select when air 3-way reject and 2-way drop-belt are used.
AIR + 2 stp DROP	Select when air 3-way reject and 2-way 2-step drop belt is used.

5.9.4 Program Number

The Program Number screen is called up by pressing the Program Number index on the Machine Parameter screen.

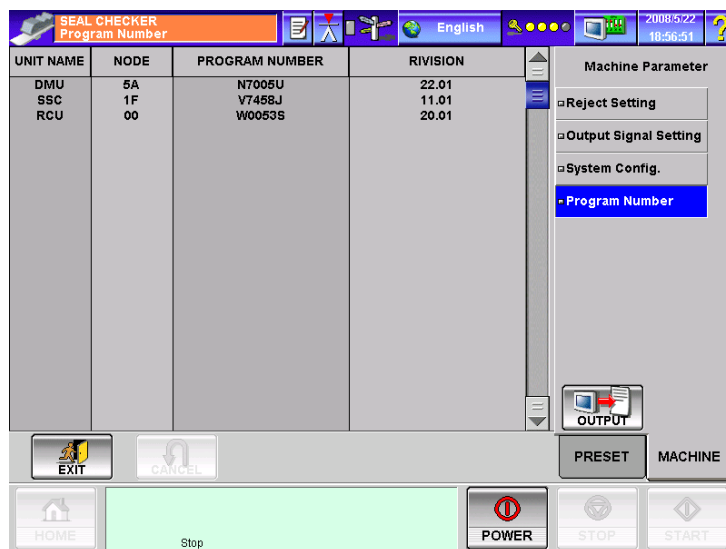


Fig.5-81 Program Number Screen

Key or Display	Function
UNIT NAME	Displays the unit name of which program has been installed.
NODE	Displays the node number of each unit in the network inside TSC-R.
PROGRAM NUMBER	Displays the number of installed software.
REVISION	Displays the version number of installed software number.

5.10 Information Screen

The Information screen is used to select / display various information on the seal checker (STATISTICS, ERROR LOG, WORST LOG, ST 16LOG, ST GRAPH).

To display the Information screen, follow the procedures below:

1. Press the [Information] key on the MAIN MENU screen.

►The Information screen appears.

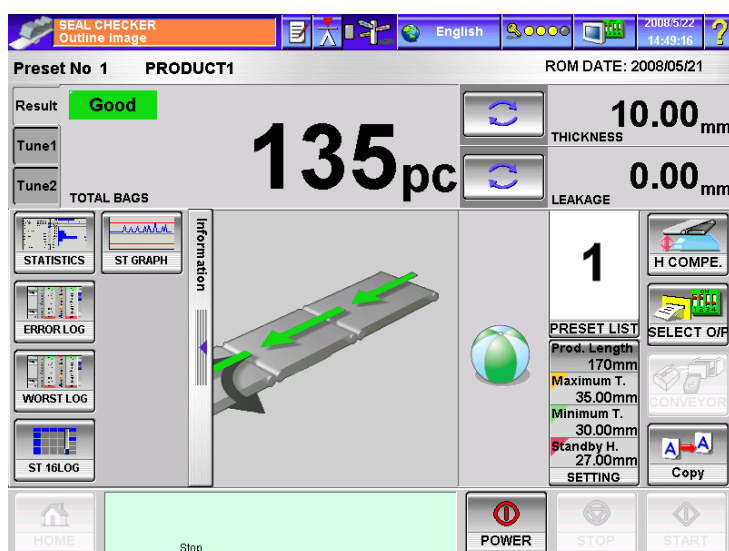
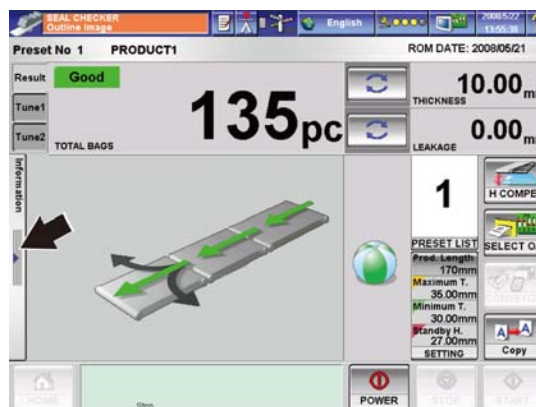


Fig.5-82 Information Screen

Key	Function
STATISTICS	Displays the STATISTICS screen (for displaying statistics). Refer to " 5.10.1 STATISTICS Screen ".
ERROR LOG	Displays the ERROR LOG screen (for displaying error history). Refer to " 5.10.2 ERROR LOG Screen ".
WORST LOG	Displays the WORST ERROR 10 screen (for displaying error statistics). Refer to " 5.10.3 WORST ERROR 10 Screen ".
ST 16LOG	Displays the Measurement Result LOG screen. Refer to " 5.10.4 Measurement Result Log Screen ".
ST GRAPH	Displays the Measurement Result Graph screen (for displaying thickness graph). Refer to " 5.10.5 Measurement Result Graph Screen ".

5.10.1 STATISTICS Screen

The STATISTICS screen is called up by pressing the [STATISTICS] key on the Information screen. The STATISTICS screen displays statistical information such as the number of products, the number of faulty products, and efficiency.

1. Press the [STATISTICS] key on the Information screen.

►The STATISTICS screen appears.

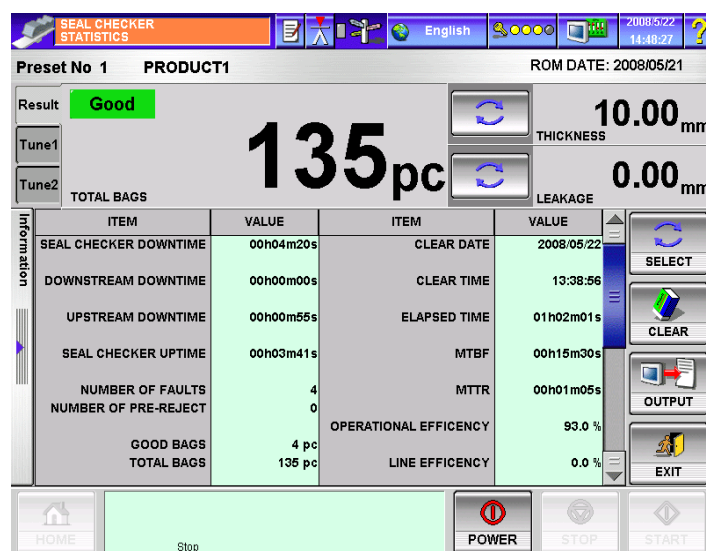


Fig.5-83 STATISTICS Screen

Key or Display	Function
SEAL CHECKER DOWNTIME	Down time of TSC-R.
DOWNSTREAM DOWNTIME	Down time of downstream equipment.
UPSTREAM DOWNTIME	Down time of upstream equipment.
SEAL CHECKER UPTIME	Operation time of TSC-R.
NUMBER OF FAULTS	Number of failure stops.
NUMBER OF PRE-REJECT	Indicates the number of pre-reject triggered. *1
GOOD BAGS	Number of bags of good product.
TOTAL BAGS	Total number of bags.
CLEAR DATE	Deletion date of statistics.

Key or Display	Function
CLEAR TIME	Deletion time of statistics.
ELAPSED TIME	Passing time.
MTBF	Mean Time Between Failure. The time till the system fails again since repaired.
MTTR	Mean Time To Repair. The time to repair the system.
OPERATIONAL EFFICENCY	Efficiency (Operating rate). $\text{Efficiency} = [\text{MTBF}] \div ([\text{MTBF}] + [\text{MTTR}])$
LINE EFFICENCY	$\text{Efficiency} = \text{Total line operating hours} \times 100 / (\text{Total line operating hours} + \text{Error stopped hours})$
THICKNESS MAX	Maximum thickness.
THICKNESS MIN	Minimum thickness.
THICKNESS AVG	Average thickness.
THICKNESS STD	Standard deviation.
GOOD BAGS	Number of good bags.
LEAK BAGS	Number of air leak bags.
FLAT BAGS	Number of flat bags.
THICK BAGS	Number of thick bags.
DOUBLE BAGS	Number of double bags.
EXT.1 N/G BAGS	Number of external 1 N/G bags.
EXT.2 N/G BAGS	Number of external 2 N/G bags.
CONT. N/G BAGS	Number of continuous N/G bags.
DOWN REJ. BAGS	Number of rejected bags at the downstream.
TOTAL BAGS	Number of total bags.
SELECT key	Switches between the number of acceptable/NG products and the ratio of acceptable/NG products to total products.
CLEAR key	Clears the statistical data of the currently selected preset No.
EXIT key	Exits from the STATISTICS screen.
OUTPUT(option) key	Prints out the statistical information.

*1: This indicates the numbers of pre-reject air valve triggered. This does not indicate “bagger manufactured - bags inspected by TSC-R.” Please use this value for your reference.

Ex.) When five (5) improper (open) bags come in a row, the pre-rejector keeps the air valve “ON”. In this case, it will be recorded as “one time”.

TIP

- MTBF (Mean Time Between Failure)
IT means how often the system fails on average.
- MTTR (Mean Time To Repair)
It means how many hours it takes for repairing the system. It also called to MTRF (Mean Time to Repair Failure).

By confirming the two values, the machine efficiency can be calculated.

$$\text{Efficiency} = [\text{MTBF}] \div ([\text{MTBF}] + [\text{MTTR}])$$

5.10.2 ERROR LOG Screen

The ERROR LOG screen is called up by pressing the [ERROR LOG] key on the Information screen. The ERROR LOG screen displays the history of the latest 100 errors.

1. Press the [ERROR LOG] key on the Information screen.

► The ERROR LOG screen appears.



SEAL CHECKER ERROR LOG				
Preset No 1 PRODUCT1		ROM DATE: 2008/05/21		
Result	Good	135pc	THICKNESS 10.00 mm	LEAKAGE 0.00 mm
Tune1				
Tune2				
TOTAL BAGS				
Information	No.	ERROR NAME	DATE & TIME	DOWN TIME
	1	Conditioner overload	2008/05/22 14:45	00h02m25s
	2	Conditioner overload	2008/05/22 14:44	00h00m39s
	3	Conditioner overload	2008/05/22 14:44	00h00m24s
	4	Jam up at Seal Checker	2008/05/22 13:41	00h00m12s
	5	Seal check conveyor speed error	2008/04/28 18:54	00h02m14s
	6	Seal check conveyor speed error	2008/04/28 18:53	00h00m56s
	7	Seal check conveyor speed error	2008/04/28 18:51	00h01m52s
	8	Seal check conveyor speed error	2008/04/28 18:51	00h00m35s
	9	Conditioner overload	2008/04/28 18:49	00h01m15s
	10			
CLEAR DATE 2008/03/31 13:14				
HOME Stop POWER STOP START				

Fig.5-84 ERROR LOG Screen

Key or Display	Function
ERROR NAME	Error name.
DATE & TIME	Occurrence date and time.
DOWN TIME	Total time of error stop.
CLEAR DATE	Error history clear date and time.
CLEAR key	Clears the error history.
OUTPUT(option) key	Prints out the history of the latest 100 errors. When errors do not reach 100, printout is executed to the number of errors occurred.
EXIT key	Exits from the LATEST 100 ERROR LOG screen.

5.10.3 WORST ERROR 10 Screen

The WORST ERROR 10 screen is called up by pressing the [WORST LOG] key on the Information screen.

The WORST ERROR 10 displays the ten errors which occur most frequently.

This screen is used for analyzing the tendency of error occurrence and obtaining the effective information for adjustment of seal checker or calibration of the setting values.

1. Press the [WORST LOG] key on the Information screen.

► The WORST ERROR 10 screen appears.



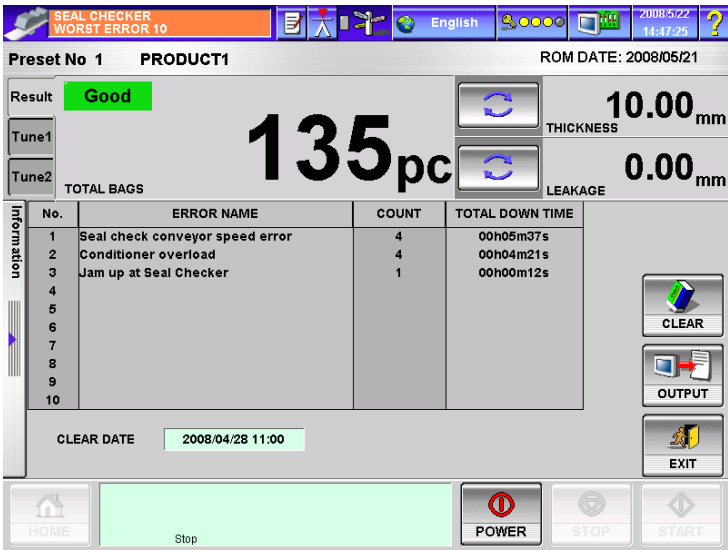


Fig.5-85 WORST ERROR 10 Screen

Key or Display	Function
ERROR NAME	Error name.
COUNT	Occurrence frequency.
TOTAL TIME	Total error down time for an applicable error.
CLEAR DATE	Error history clear date and time.
CLEAR key	Clears the error history.
OUTPUT key	Prints out the displayed error history.
EXIT key	Exits from the WORST ERROR 10 Log screen.

5.10.4 Measurement Result Log Screen

The Measurement Result Log screen is called up by pressing the [ST 16LOG] key on the Information screen.

The Measurement Result Log screen displays the measurement result of the latest 16 seal check result, air leakage, thickness and interval.

This screen is used for confirming the parameters of the upstream equipment and whether the data is properly set.

- Press the [ST 16LOG] key on the Information screen.

►The Measurement Result Log screen appears.



SEAL CHECKER Measure Result Log				
Preset No 1 PRODUCT1		ROM DATE: 2008/05/21		
Result	Good		THICKNESS	10.00 mm
Tune1	TOTAL BAGS		LEAKAGE	0.00 mm
Tune2				
Information	No.	RESULT	LEAKAGE [mm]	THICKNESS [mm]
	1			
	2			
	3			
	4			
	5			
	6			
	7			
	8			
	9			
	10			
	11			
	12			
	13	Good	0.00	15.12
	14	Good	0.00	15.12
	15	Good	0.00	15.12
	16	Good	0.00	15.12
				INTERVAL [msec]
				65535
				652
				559
				582
HOME Stop POWER STOP START				

Fig.5-86 Measurement Result Log Screen

Key or Display	Function
RESULT	Seal check result.
LEAKAGE	Air leakage measurement.
THICKNESS	Thickness measurement.
INTERVAL	Interval measurement.
OUTPUT key	Prints out the measurement result log.
EXIT key	Exits from the Measurement Result Log screen.

5.10.5 Measurement Result Graph Screen

The Measurement Graph Log screen is called up by pressing the [ST GRAPH] key on the Information screen.

The Measurement Result Graph screen displays the change of product thickness passing through the seal checker.

- 1. Press the [ST GRAGH] key on the Information screen.
 - The Measurement Result Graph screen appears.

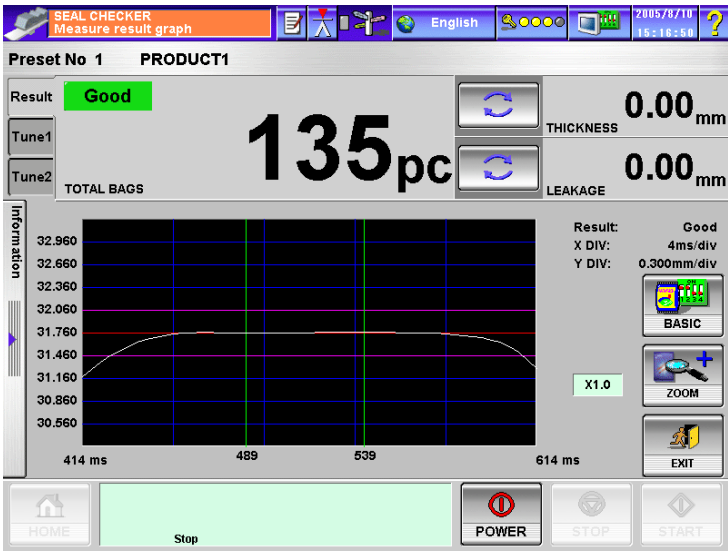
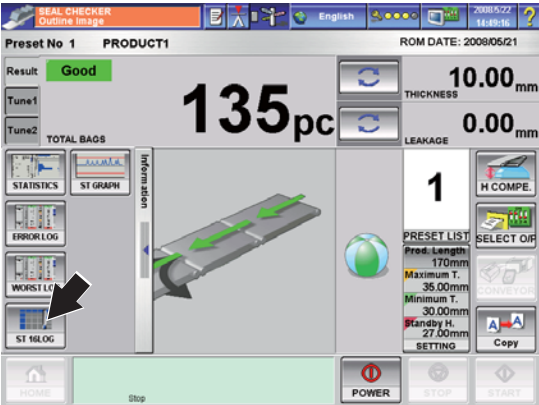


Fig.5-87 Measurement Result Graph Screen

Key or Display	Function
Result	Seal check result.
X DIV.	Scale interval on the horizontal axis (X-axis).(ms/div)
Y DIV.	Scale interval on the vertical axis (Y-axis).(mm/div)
BASIC key	Changes the Basic Setting screen.
ZOOM key	Switches the scale of graduation for the columns.
EXIT key	Exits from the Measurement Result Graph screen.

5.11 Production Screen

The Production screen is called up by pressing the [START] key in the MAIN MENU screen. This is the standard screen that appears during operation.

During operation, the user can call up the Result, Tune1 and Tune2 screens individually by pressing the relevant tabs.

The Result screen displays the data related to the parameters. It also displays the functions currently available.

The operation screen displays the total number of products, measured value of thickness, and measured value of air leakage.

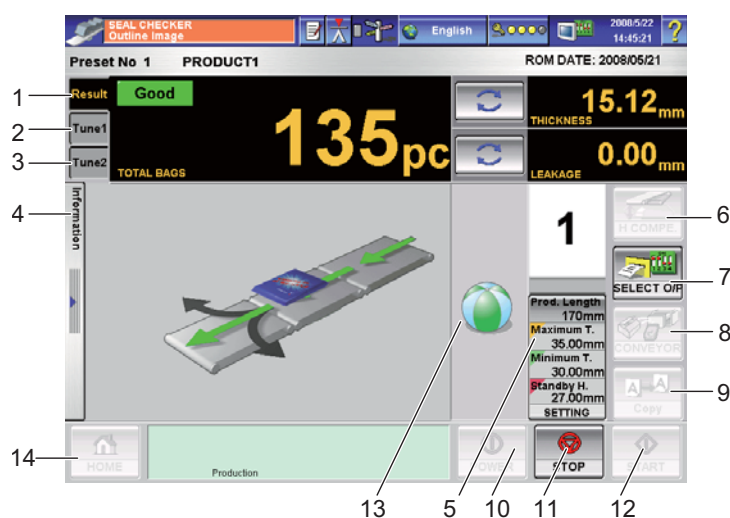


Fig.5-88 Production Screen

No.	Key or Display	Function
1	Result Tab	Displays the inspection result. Good: Good bag LEAK: Air leak bag THICK: Thick bag THIN: Thin bag Refer to " 5.4.2.1 Result ".
2	Tune1 Tab	Refer to " 5.4.2.2 Tune1 ".
3	Tune2 Tab	Refer to " 5.4.2.3 Tune2 ".
4	Information	Displays the INFORMATION screen for statistics, error log, etc. Refer to " 5.10 Information Screen ".
5	SETTING key	Displays the Preset Parameter screen. The major set values are displayed above the key. Refer to " 5.8 PRESET PARAMETER".
6	H COMPE. key	Executes Height Compensation. (Refer to "8.3.4" Height Compensation of Seal Check Conveyor") This key is invalid when the machine is in production.
7	SELECT O/P key	Displays the Output Select Screen. Refer to " 5.7 Output Selection".

No.	Key or Display	Function
8	CONVEYOR key	Discharges products on the conveyor. This function is valid when the power is ON and the system is stopped. This key is invalid when the machine is in production.
9	COPY key	Displays the Copy screen. (Refer to “5.6 Preset Copy”) This key is invalid when the machine is in production.
10	POWER key	Turns ON the drive power of the system. When the power is ON, the key lights up. This function is invalid during operation.
11	STOP key	Turns OFF the stop the system.
12	START key	Press this to start the system while the POWER key is ON. Press this to stop the system while the START key is ON.
13		Good bag
		Air leak bag
		Thick bag
		Thin bag
14	HOME(option)	Displays the LINE CONTROL HOME MENU Screen. (applied with ITPS)

5.12 Automated Adjustment Unit

5.12.1 Normal Operation

Set the stabilizer width and incline conveyor height automatically to the entered amount.

Each section has measure and pointer to help width/height adjustment.

The setting parameters are following two (2) items. These can set in General Setting tub in Basic Setting.

Name	Detail
Bag Stabilizer Width	Set the width of stabilizer brushes.
Infeed Conv. Height	Set the bag length in inches. The height will be adjusted automatically to the location where a bag with input length can be handled without problem. *1

*1 : The pointer on the incline conveyor does not refer “concrete height”. When the pointer indicates “11” on the measure, this means that the conveyor height is set to the height where 11-inch long bag can be handled properly. When “6” is indicated, conveyor height is set sufficient to handle 6-inch long bag properly. Smaller number comes to upper location.

Automated adjustment unit operates on the following condition:

Name	Detail
Servo ON Completed	When the TSC-R’s servo motor becomes ON, automated adjustment unit performs zero adjustment, then moves to the setting location. The zero locations are lowest position of the infeed conveyor and the widest location of the stabilizer brushes. *1 *2 *3
Servo on/Operation	When setting parameter or preset numbers are changed, automated adjustment unit moves to the set position. *2 *3
Servo OFF	Automated adjustment unit goes back to zero position before becoming servo OFF.

*1 : The zero (default) locations are lowest position of the infeed conveyor and the widest location of the stabilizer brushes.

*2 : When turning on the main power, please make sure there are no obstacles around the machine, and be careful for the moving parts.

*3 : The operation time for automated adjustment unit is approx. 1 min. at longest; but depends on current setting and unit location upon servo ON/OFF.

5.13 ATA Sampling (Optional)

To do ATA, twenty (20) product samples will be collected. When sampling is completed, TSC-R sends an ATA completion signal for one (1) second to upstream equipment. ATA sampling is available only for the products that are rejected by drop belt reject.

5.13.1 ATA Procedure

1. Press the [ATA] key on the main screen.

- ▶ ATA button grays out unless the product is rejected by drop belt reject.



Fig.5-89 Main Screen

- ▶ While TSC is preparing for ATA, a progress bar is indicated on the display for fifteen (15) seconds (ATA preparation). During ATA preparation;

- TSC keeps normal operation.
- Improper items will be rejected when detected.
- TSC goes back to normal operation if [Cancel] key is pressed.

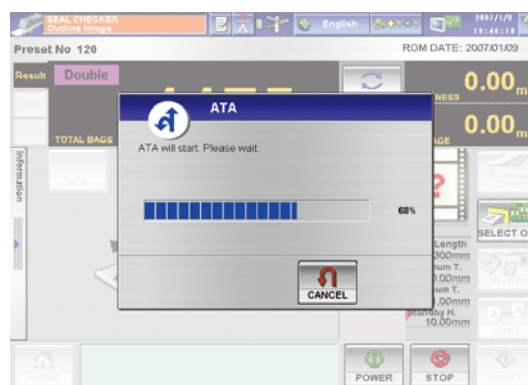


Fig.5-90 ATA Preparation Screen

- ▶ The very first item after finishing ATA preparation will be the trigger to start ATA sampling. Progress bar disappears and the message on the screen changes after detecting this first item. On this condition;

- The very first item (trigger item) is rejected if it is improper item.
- When TSC detects the next item to the trigger bag, the drop belt goes down for sampling.
- Drop belt remain at down position until twenty (20) bags are collected.
- If [Cancel] key is pressed, the unit goes back to normal operation.

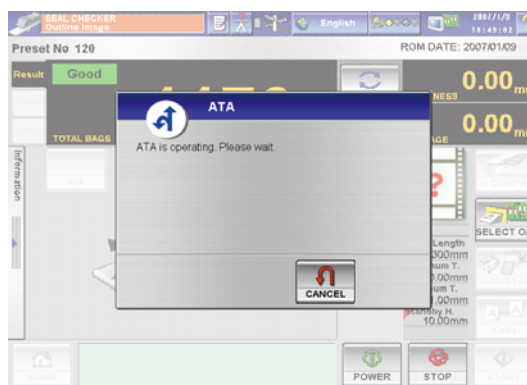


Fig.5-91 ATA Sampling screen

► When twenty (20) bag sampling is completed, there shows “ATA is Completed!!!” on the screen. At the same time, ATA completion signal is sent to upstream equipment. Please confirm that the ATA completion screen is displayed, unless ATA completion signal will not be sent.

2. Press the [OK] key on the confirmation screen.

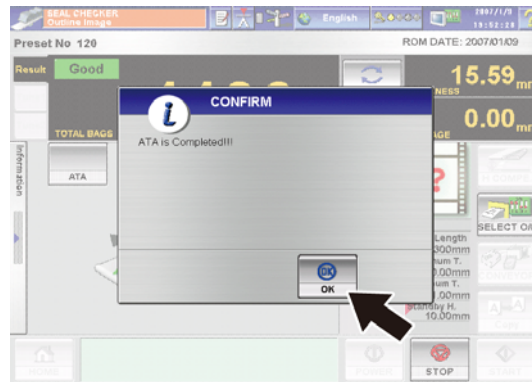


Fig.5-92 Confirmation Screen (ATA completed)

5.13.2 Attention

- (1) ATA sampling cannot be done unless the reject method is drop belt.
- (2) The following parameters are fixed and cannot be changed:
 - ATA Preparation time 15 seconds
 - Numbers of ATA sampling 20 bags
 - ATA completion signal ON time 1 second
- (3) Items passed through during ATA sampling will be referred to statistical data.
- (4) If an error occurs during ATA process, ATA sampling will be automatically finished. Please release the error then again press ATA button manually to re-start ATA sampling.
- (5) During ATA sampling, ALL the items will be sampled regardless of good or bad bags. In case open bags or scattered products conveyed, there may be a difference in numbers between actually sampled product and counted ones by TSC.